PHYSICS

1) 1 mole O_2 gas and 1 mole N_2 gas are mixed in an enclosure of volume 5 litre and temperature 27° C. Find pressure of mixture :-

(1) 5 atm

(2) 15 atm

(3) 1 atm

(4) 10 atm

2) A diatomic gas expands in a process according to law P^3V^2 = const. Then, find molar specific heat of gas in this process.

(1) $\frac{13}{2}R$

(2) $\frac{7}{2}R$

(3) $\frac{15}{2}R$

(4) $\frac{11}{2}R$

@examlix

3) Two walls of thicknesses d_1 and d_2 and thermal conductivity k_1 and k_2 are in contact. In the steady state, if the temperatures at the outer surface are T_1 and T_2, the temperature at the common wall is

(1) $\frac{K_1T_1d_2 + K_2T_2d_1}{K_1d_2 + K_2d_1}$

(2) $\frac{K_1T_1 + K_2T_2}{d_1 + d_2}$

(3) $\left(\frac{K_1d_1 + K_2d_2}{T_1 + T_2}\right)$

(4) $\frac{K_1d_1T_1 + K_2d_2T_2}{k_1d_1 + k_2d_2}$

4) Cooling rate of a sphere of 600 K at external environment (200K) is R. When the temperature of sphere is reduced to 400 K then cooling rate of the sphere becomes :

(1) $\frac{3}{16}R$

(2) $\frac{16}{3}R$

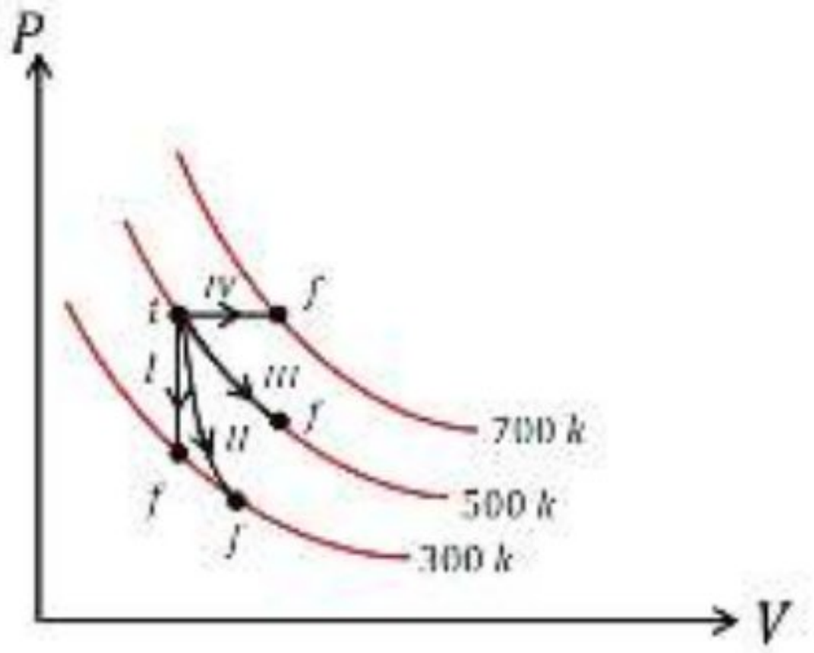
(3) $\frac{9}{27}R$

(4) None

Watermarkly

Scanned with OKEN Scanner

5) Thermodynamic processes are indicated in the following diagram



Match the following:

Column-I	Column-II
P. Process I	a. Adiabatic
Q. Process II	b. Isobaric
R. Process III	c. Isochoric
S. Process IV	d. Isothermal

(1) P-d, Q-b, R-a, S-c

(2) P-a, Q-c, R-d, S-b

(3) P-c, Q-a, R-d, S-b

(4) P-c, Q-d, R-b, S-a

6) 50 gram of ice at 0°C is mixed with 50 gram of water at 60°C, final temperature of mixture will be :-

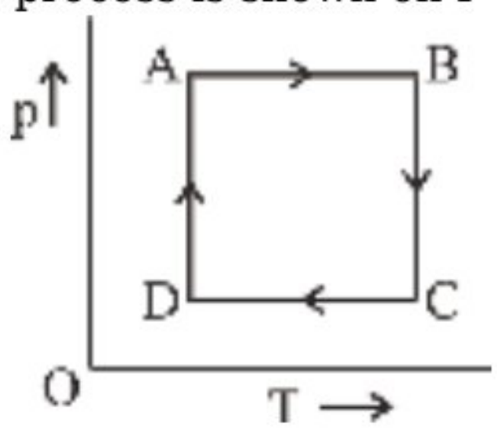
(1) 15°C

(2) 0°C

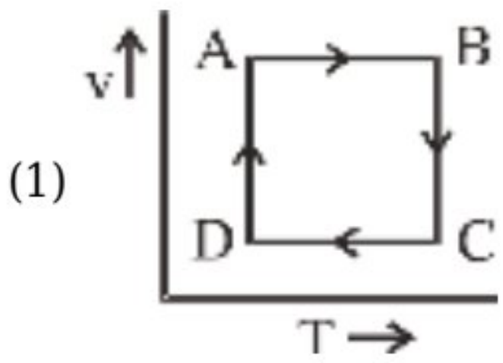
(3) 10°C

(4) 40°C

7) A cyclic process is shown on P-T diagram. Which of the curve shows the same process on a V-T diagram :-



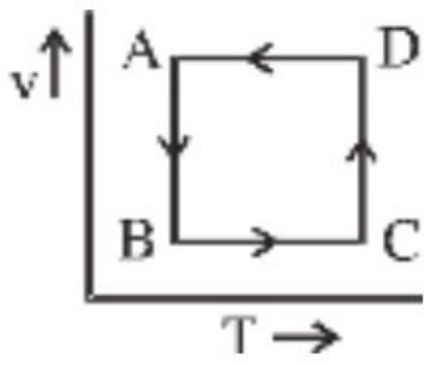
(1)



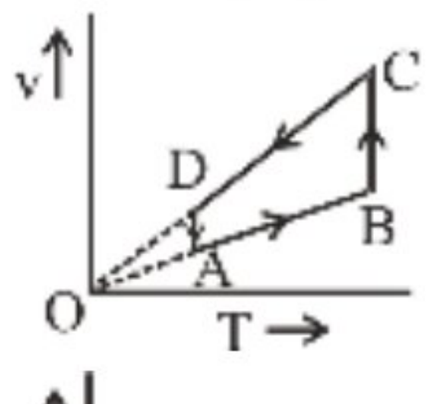
@examlix
Watermarkly
Scanned with OKEN Scanner

NOVA

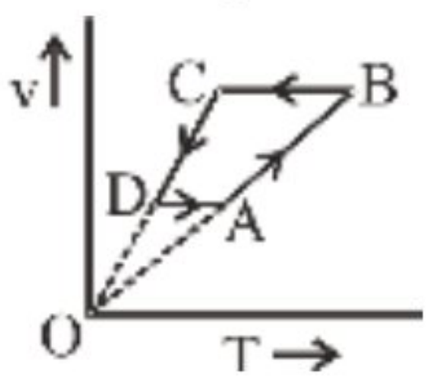
(2)



(3)



(4)



8) The ratio of specific heats (γ) of an ideal gas is given :-

(1) $\frac{1}{1-\left(\frac{R}{C_p}\right)}$

(2) $1+\frac{R}{C_v}$

(3) $\frac{C_p}{(C_p-R)}$

(4) All of these

@examlix

9) Two star emit maximum radiation of wavelength 3600 Å and 4800 Å respectively then find ratio of their temperature is –

(1) 1 : 2

(2) 3 : 4

(3) 4 : 3

(4) 2 : 1

10) On increasing the temperature of a gas filled in a closed container by 1°C its pressure increases by 0.4%, then initial temperature of the gas is -

(1) 25°C

(2) 250°C

(3) 250 K

(4) 2500°C

11) A body takes 4 minutes to cool from 100°C to 70°C. To cool from 70°C to 40°C it will take- (room temperature is 15°C)

Watermarkly

Scanned with OKEN Scanner

(1) 7 minutes

(2) 6 minutes

(3) 5 minutes

(4) 4 minutes

12) At what temperature will the oxygen molecules have the same root mean square speed as hydrogen molecules at 200 K-

(1) 800 K

(2) 1600 K

(3) 2400 K

(4) 3200 K

13) The temperature gradient in a rod of 0.5 m long is 80°C/m. If the temperature of hotter end of the rod is 30°C, then the temperature of the cooler end is :-

(1) 40°C

(2) –10°C

(3) 10°C

(4) 20°C

14) Heat is transmitted from higher to lower temperature through actual mass motion of the molecules in:-

(1) Conduction

(2) Convection

(3) Radiation

(4) All of above

15) A black body is at a temperature of 2880 K. The energy of radiation emitted by the body at wavelength 500 nm is U_1, at wavelength 1000 nm is U_2 and that at 2000 nm is U_3. Wien's constant, b = 2.88×10^6 nmK. Which of the following is correct ?

(1) $U_1 = 0$

(2) $U_3 = 0$

(3) $U_1 > U_2$

(4) $U_2 > U_1$

16) The molar heat capacity for an ideal gas - (a) is zero for adiabatic process.
(b) is infinite for an isothermal process
(c) depends on the nature of gas for a process in which either volume or pressure is constant
(d) is equal to the product of the molecular weight and gram specific heat capacity for any process.
Choose the correct statements :-

(1) a, b

(2) c,d

@examlix

Watermarkly

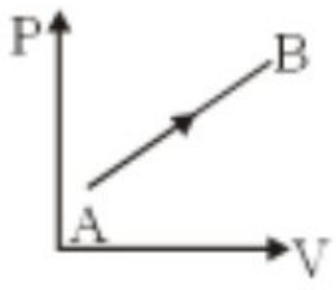
Scanned with OKEN Scanner

(3) a, b, c & d

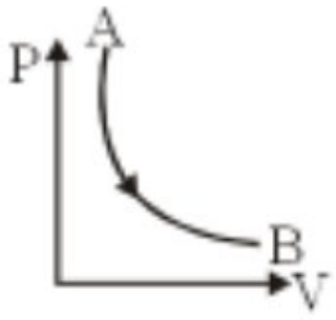
(4) b, c, d

17) The process $\Delta U = 0$, for an ideal gas can be best represented in the form of a graph :-

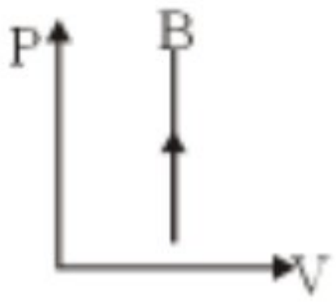
(1)



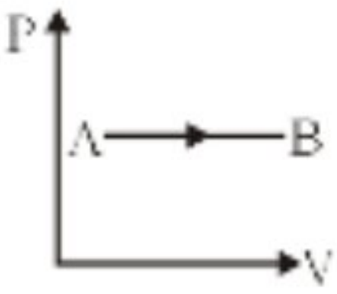
(2)



(3)



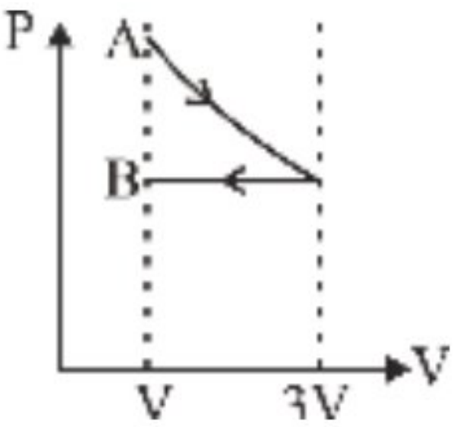
(4)



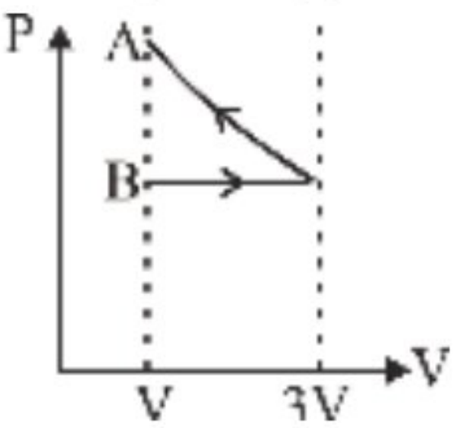
@examlix

18) One mole of an ideal gas goes from an initial state A to final state B via two processes: It first undergoes isothermal expansion from volume V to 3V and then its volume is reduced from 3V to V at constant pressure. The correct P-V diagram representing the two processes is :-

(1)



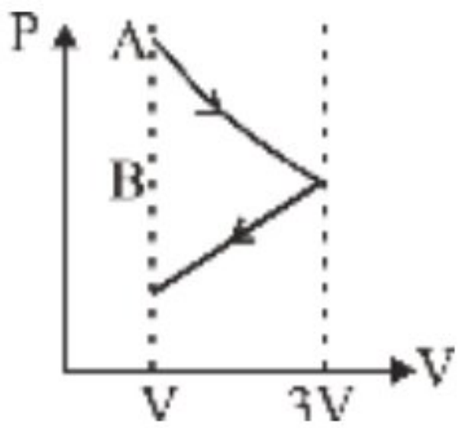
(2)



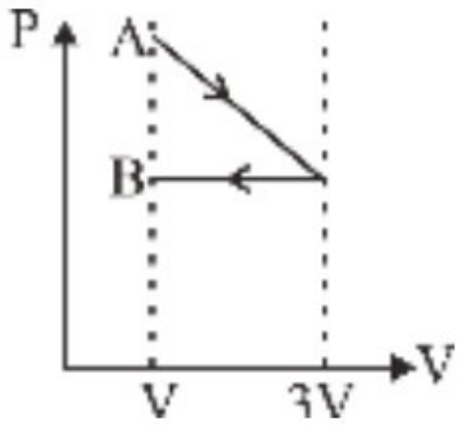
Watermarkly

Scanned with OKEN Scanner

(3)



(4)



19) If water is heated on top of Mount Everest, where atmospheric pressure is very low, water will start boiling at temp : -

(1) At temperature less than 100°C

(2) At fixed 100°C

(3) At temperature more than 100°C

(4) Water will freeze on heating

20) Two rods one of aluminium of length ℓ_1 having coefficient of linear expansion α_a , and other steel of length ℓ_2 having coefficient of linear expansion α_s are joined end to end. The expansion in both the rods is same on variation of temperature. Then the value of $\frac{\ell_1}{\ell_1 + \ell_2}$ is

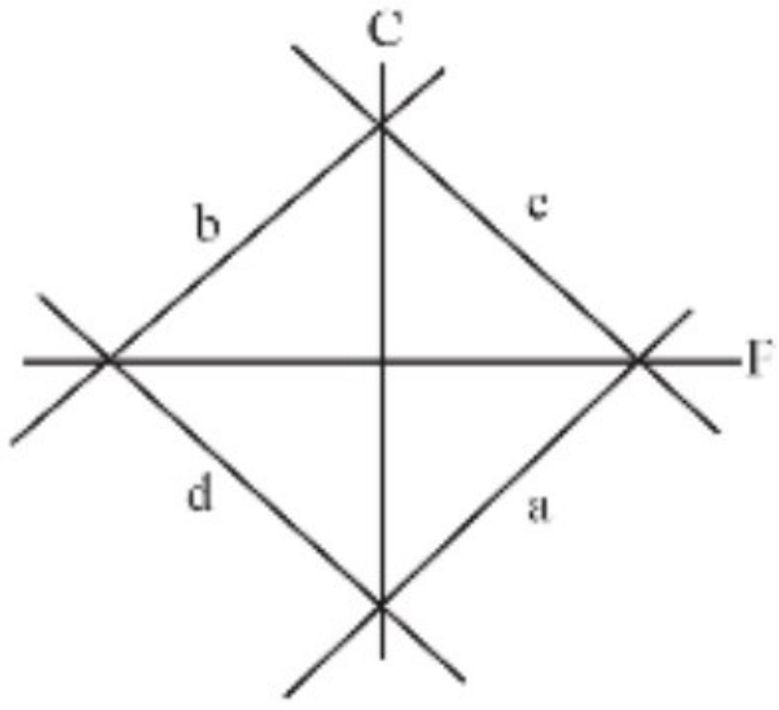
(1) $\frac{\alpha_s}{\alpha_a + \alpha_s}$

(2) $\frac{\alpha_s}{\alpha_a - \alpha_s}$

(3) $\frac{\alpha_a + \alpha_s}{\alpha_s}$

(4) None of these

21) Which of the curves in figure represents the relation between Celsius and Fahrenheit temperature?



@examlix

Watermarkly

Scanned with OKEN Scanner

(1) Curve a

(2) Curve b

(3) Curve c

(4) Curve d

22) Which of the following is correct according to Claussius statement of second law of thermodynamics.

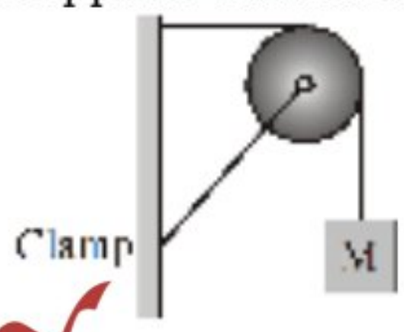
(1) It is possible to transfer heat from a low temp. reservoir to a high temp. reservoir.

(2) No heat engine can have 100% efficiency.

(3) Moving heat from a cold region to a warm region requires an external energy or work input.

(4) None of these

23) A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given by :



(1) $\sqrt{2}$ mg

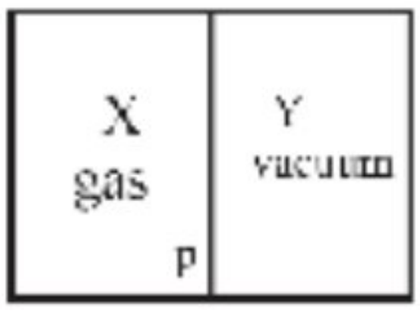
(2) $\sqrt{2}$ Mg

(3) $\sqrt{(M+m)^2+m^2g}$

(4) $g\sqrt{(M+m)^2+M^2}$

@examlix

24) A closed container is fully insulated from outside. One half of it is filled with an ideal gas X separated by a plate P from the other half Y which contains a vacuum as shown in figure. When P is removed, X moves into Y. Which of the following statements is correct?



(1) No work is done by X

(2) X decreases in temperature

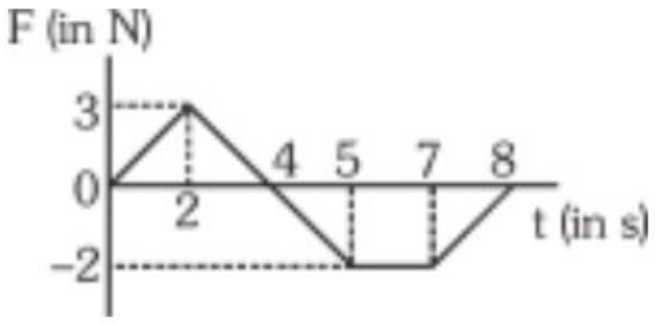
(3) X increases in internal energy

(4) X doubles in pressure

25) Force-time graph for the motion of a body is shown in fig. Change in linear momentum between 0 to 8 s is :-

Watermarkly

Scanned with OKEN Scanner



(1) Zero

(2) 4 N-s

(3) 8 N-s

(4) 6 N-s

26) A rocket of mass 60 kg. is fired in a gravity free space is ejecting gases with velocity 300 m/s at the rate of 2 kg/s. What will be the initial acceleration of the rocket ?

(1) 30 m/s^2

(2) 5 m/s^2

(3) 10 m/s^2

(4) 15 m/s^2

27) A particle moves in x-y plane under the influence of a force such that its linear momentum $\vec{P}(t) = A\left[\hat{i}\cos(kt) - \hat{j}\sin(kt)\right]$ where A and k are constants. The angle between the force and the momentum is :-

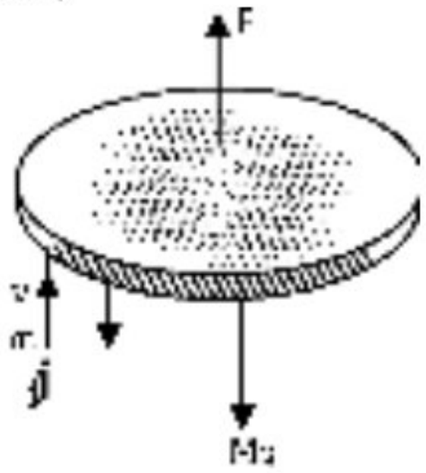
(1) 0°

(2) 30°

(3) 45°

(4) 90°

28) A disc of mass 1·0 kg kept floating horizontally in air by firing bullets of mass 0·05 kg each vertically at it, at the rate of 10 per second. If the bullets rebound with the same speed, the speed with which these are fired will be– (g = 9.8 m/s^2)



(1) 0·098 m/s

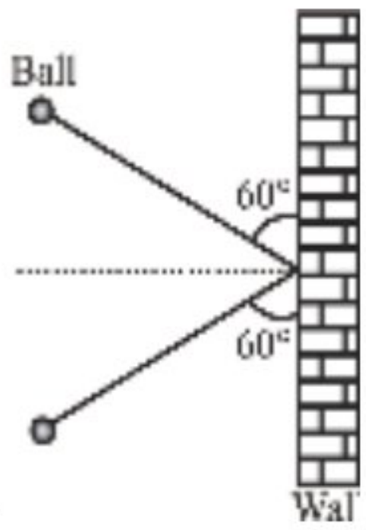
(2) 0·98 m/s

(3) 9·8 m/s

(4) 98·0 m/s

29) A ball of mass 3 kg, moving with a speed of 100 m/s, strikes a wall at an angle 60° (as shown in figure). The ball rebounds at the same speed and remains in contact with the wall for 0.2 sec; the

@examlix
Watermarkly
Scanned with OKEN Scanner



force exerted by the ball on the wall is :-

(1) $1500\sqrt{3}$ N

(2) 1500 N

(3) $300\sqrt{3}$ N

(4) 300 N

30)

The force exerted by the floor of an elevator on the foot for a person standing there, is more than the weight of the person, if the elevator is :
A. Going up with increasing speed
B. Going down with decreasing speed

(1) A is true but B is false

(2) A is false but B is true

(3) Both A and B are true

(4) Both A and B are false

@examlix

31) A rocket of initial mass 1500 kg. ejects gas at a constant rate of 10kg/s. with a relative speed of 5 km/s. What is the acceleration of the rocket 50 seconds after the blast, neglecting gravity :-

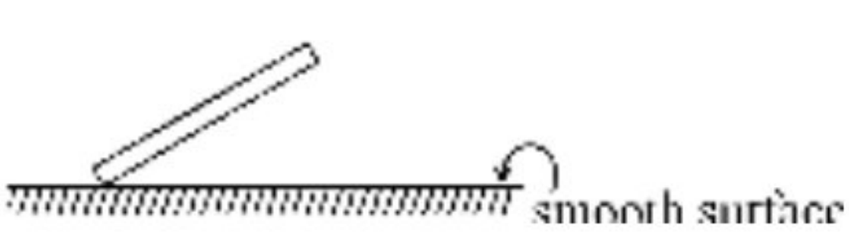
(1) 10 ms^{-2}

(2) 25 ms^{-2}

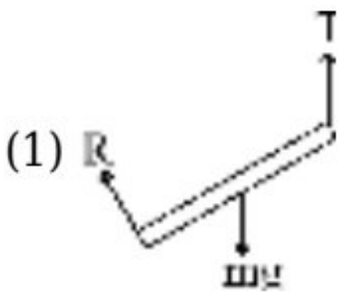
(3) 50 ms^{-2}

(4) 100 ms^{-2}

32) Which figure represents the correct F.B.D. of rod of mass m as shown in figure :

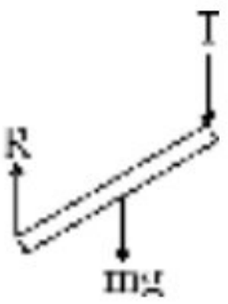


(1)



Watermarkly

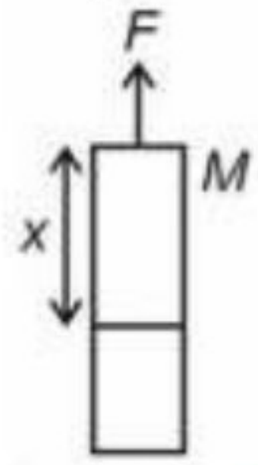
Scanned with OKEN Scanner

(2) 

(3) 

(4) None of these

33) A uniform bar of mass M and length ℓ is carried upward by applying force F on its upper end.



The force at the distance x from its upper end is

(1) $\frac{F(\ell - x)}{\ell}$

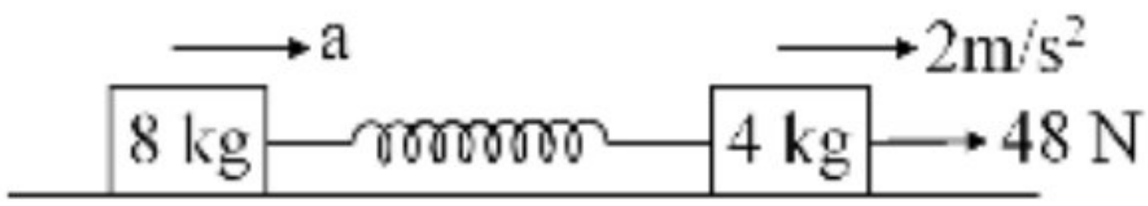
(2) $\frac{Fx}{\ell}$

(3) $\frac{(F - Mg)x}{\ell}$

(4) $\frac{(F - Mg)(\ell - x)}{\ell}$

@examlix

34) Find acceleration of 8 kg block when acceleration of 4 kg block is 2 m/s^2 in given situation –



(1) 2 m/s^2

(2) 8 m/s^2

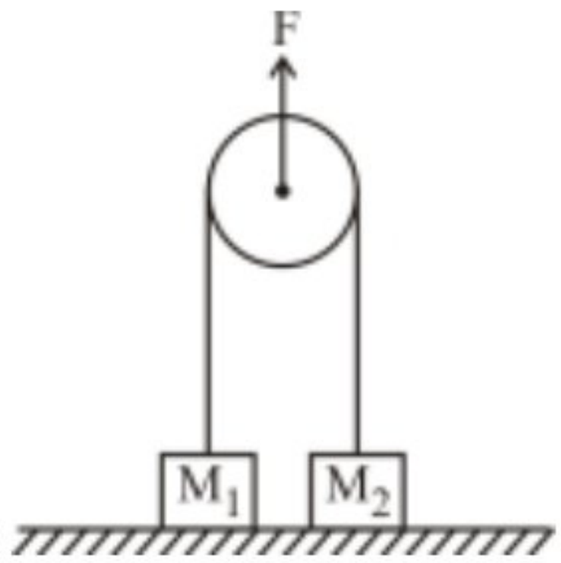
(3) 5 m/s^2

(4) 4 m/s^2

35) In shown figure M_1 = 4kg and M_2 = 8kg and the pulley is moveable. At t = 0, both masses touch the ground and the string is taut. A vertically upward force F = 2t is applied to the pulley. The time

Watermarkly

Scanned with OKEN Scanner



when M_2 is lifted off the ground is :- (take g = 10 m/s^2)

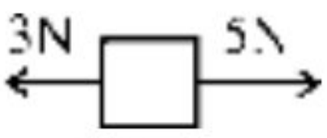
(1) 40 sec

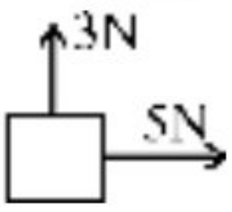
(2) 80 sec

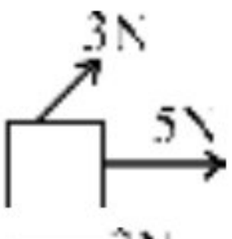
(3) 120 sec

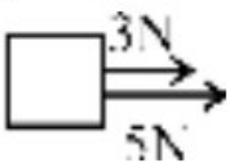
(4) 160 sec

36) Two forces, one with a magnitude of 3N and the other with a magnitude of 5N, are applied to an object. For which orientations of the forces shown in the diagrams is the magnitude of the acceleration of the object the least ?

(1) 

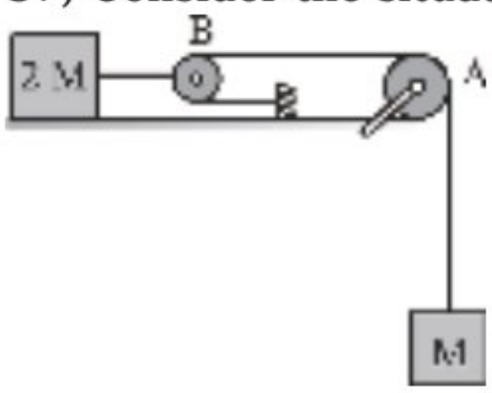
(2) 

(3) 

(4) 

@examlix

37) Consider the situation as shown in the figure:-



(1) The acceleration of M is 2g/3

(2) The tension in the string connecting A and B is Mg/3

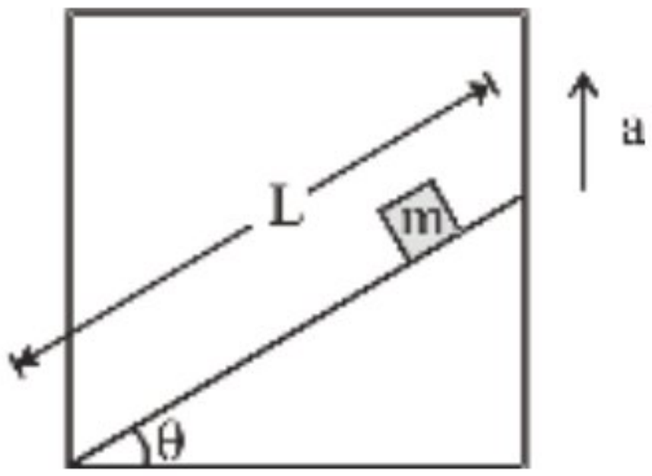
(3) The reaction on pulley A is $\sqrt{2}$ Mg/3 at an angle 45° with the horizontal

(4) All the options are correct

38) A block is sliding along an inclined plane as shown in figure. If the acceleration of chamber is a as shown in the figure. The time required to cover a distance L along incline is :-

Watermarkly

Scanned with OKEN Scanner



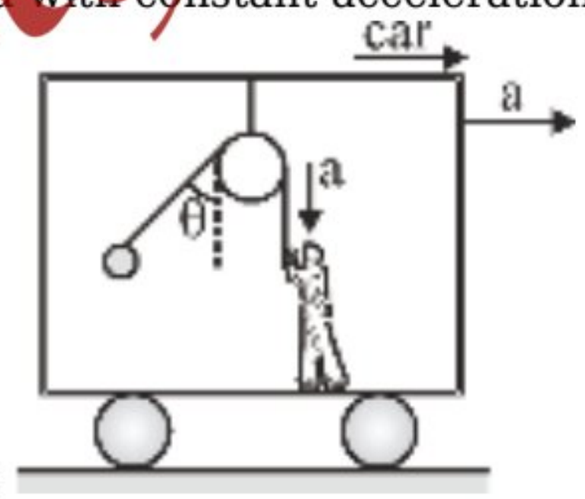
(1) $\sqrt{\frac{2L}{g\sin\theta - a\cos\theta}}$

(2) $\sqrt{\frac{2L}{g\sin\theta + a\sin\theta}}$

(3) $\sqrt{\frac{2L}{g\sin\theta + a\cos\theta}}$

(4) $\sqrt{\frac{2L}{g\sin\theta}}$

39) A bob is hanging over a pulley inside a car through a string. The second end of the string is in the hand of a person standing in the car. The car is moving with constant acceleration 'a' directed horizontally as shown in figure. Other end of the string is pulled with constant acceleration 'a'



(relative to car) vertically. The tension in the string is equal to :

(1) $m\sqrt{g^2 + a^2}$

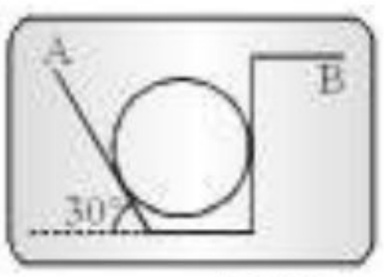
(2) $m\sqrt{g^2 + a^2} - ma$

(3) $m\sqrt{g^2 + a^2} + ma$

(4) m (g+a)

40)

The 50kg homogeneous smooth sphere rests on the 30° incline A and bears against the smooth vertical wall B. Calculate the contact forces at A and B :



(1) $N_B = \frac{1000}{\sqrt{3}}N, N_A = \frac{500}{\sqrt{3}}N$

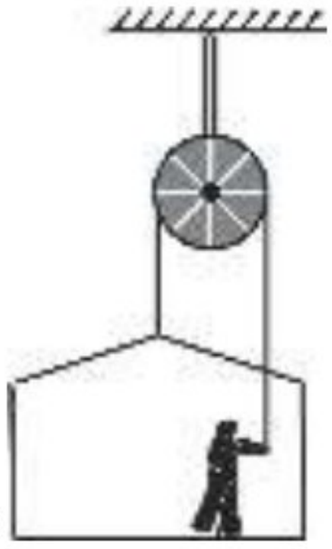
@examlix
Watermarkly
Scanned with OKEN Scanner

(2) $N_A = \frac{1000}{\sqrt{3}}N, N_B = \frac{500}{\sqrt{3}}N$

(3) $N_A = \frac{100}{\sqrt{3}}N, N_B = \frac{500}{\sqrt{3}}N$

(4) $N_A = \frac{1000}{\sqrt{3}}N, N_B = \frac{50}{\sqrt{3}}N$

41) A man is raising himself and the crate on which he stands with an acceleration of 5 ms^{-2} by a massless rope–and–pulley arrangement. Mass of the man is 100 kg and that of the crate is 50 kg. If g = 10 ms^{-2}, the tension in the rope is



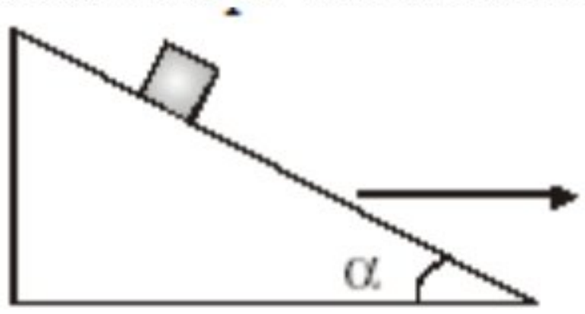
(1) 2250 N

(2) 1125 N

(3) 750 N

(4) 375 N

42) A block is kept on a frictionless inclined surface with angle of inclination 'α'. The incline is given an acceleration 'a' to keep the block stationary. Then a is equal to



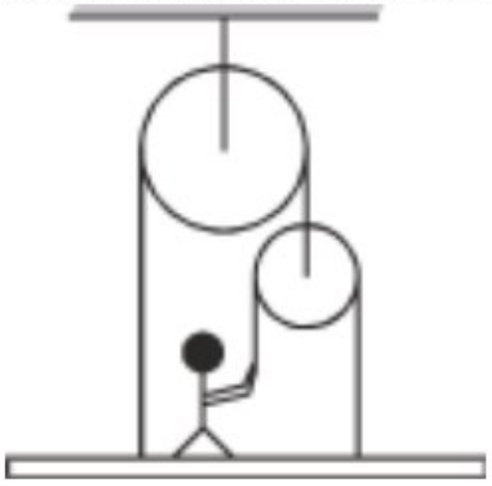
(1) g

(2) g tan α

(3) g / tan α

(4) g cosec α

43) In the given diagram, with what force must the man pull the rope to hold the plank in position? Mass of the man is 80 kg. Neglect the weights of plank, rope and pulley. Take g = 10 ms^{-2}



(1) 200 N

@examlix

Watermarkly

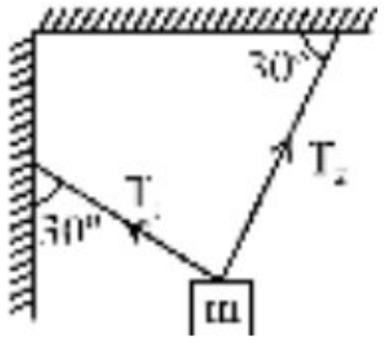
Scanned with OKEN Scanner

(2) 300 N

(3) 600 N

(4) 150 N

44) Calculate T_1 & T_2 :-



(1) $\frac{\sqrt{3}}{2}mg, \frac{mg}{2}$

(2) $mg, \frac{\sqrt{3}}{2}mg$

(3) $\frac{\sqrt{3}}{2}mg, \frac{\sqrt{3}}{2}mg$

(4) $\frac{mg}{2}, \frac{mg}{2}$

45) A vessel has 6g of oxygen at pressure P and temperature 400K. A small hole is made in it so that oxygen leaks out. How much oxygen leaks out if the final pressure is P/2 and temperature is 300 K?

(1) 3g

(2) 2g

(3) 4g

(4) 5g

@examlix

CHEMISTRY

1) If 1 mL of water contains 20 drops. Then number of molecules in a drop of water is :-

(1) 6.02×10^{23}

(2) 1.67×10^{21}

(3) 1.34×10^{18}

(4) 4.34×10^{20}

2) If 10^{21} molecules are removed from 200 mg of CO_2, then the number of moles of CO_2 left is :-

(1) 2.88×10^{-3}

(2) 1.66×10^{-3}

(3) 4.54×10^{-3}

(4) 1.66×10^{-2}

3) How many moles of magnesium phosphate, $Mg_3(PO_4)_2$ will contain 0.25 mole of oxygen atoms?

Watermarkly

Scanned with OKEN Scanner

(1) 3.125×10^{-2}

(2) 1.25×10^{-2}

(3) 2.5×10^{-2}

(4) 0.02

4) How many moles of electron weigh one -kilogram ?

(1) 6.023×10^{23}

(2) $\frac{1}{9.108} \times 10^{31}$

(3) $\frac{6.023}{9.108} \times 10^{54}$

(4) $\frac{1}{9.108 \times 6.023} \times 10^{8}$

5) 112.0 mL of NO_2 at STP (1atm, 273K) was liquefied, the density of the liquid being 1.15 g mL^{-1}. Calculate the volume and the number of molecules in the liquid NO_2(At. wt. N = 14)

(1) 0.10 mL and 3.01×10^{22}

(2) 0.20 mL and 3.01×10^{21}

(3) 0.20 mL and 6.02×10^{23}

(4) 0.40 mL and 6.02×10^{21}

6) Elements A and B form two compounds B_2A_3 and B_2A. 0.05 moles of B_2A_3 weight 9.0 gms and 0.10 mole of B_2A weight 10 gms atomic weight of A and B are-

(1) 20 and 30

(2) 30 and 40

(3) 40 and 30

(4) 30 and 20

7)

For the reaction,
$2Fe(NO_3)_3 + 3Na_2CO_3 \rightarrow Fe_2(CO_3)_3 + 6NaNO_3$
Initially if 2.5 mole of $Fe(NO_3)_2$ and 3.6 mole of Na_2CO_3 is taken. If 6.3 mole of $NaNO_3$ is obtained then % yield of given reaction is:

(1) 50

(2) 84

(3) 87.5

(4) 100

8) $CaCO_3$ is 90% pure. Volume of CO_2 collected at STP when 10 g of $CaCO_3$ is decomposed is -

(1) 2.016 L

@examlix

Watermarkly

Scanned with OKEN Scanner

NOVA

(2) 1.008 L

(3) 10.08 L

(4) 20.16 L

9) Two oxides of a metal contain 22.22% and 30% oxygen by mass respectively. If the formula of the first oxide is MO, then the formula of the second oxide is :-

(1) MO

(2) MO_2

(3) M_2O_3

(4) M_2O

10) The **right** option for the mass of CO_2 produced by heating 20 g of 20% pure limestone is (Atomic mass of Ca = 40)

[$CaCO_3 \xrightarrow{1200\ K} CaO + CO_2$]

(1) 1.76 g

(2) 2.64 g

(3) 1.32 g

(4) 1.12 g

11) The molecular formula of anthracene  How many structural isomers are possible when one of the hydrogen is replaced by a chlorine atom ?

@examlix

(1) 6

(2) 3

(3) 4

(4) 2

12) The total number of benzene derivatives with molecular formula $C_6H_3Cl_3$ is :

(1) 2

(2) 3

(3) 4

(4) 5

13) An alkane of molecular weight 114 gives only one monochlorogenated product. The alkane is :-

(1) 2,2,3,3-Tetra methyl butane

(2) 2-Methyl hexane

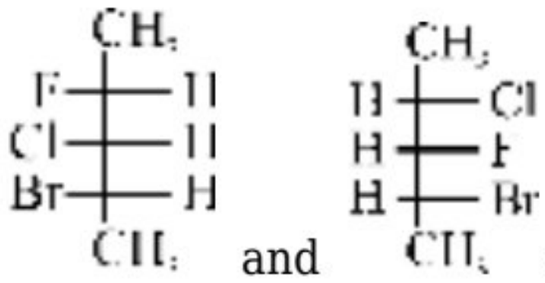
(3) 2,2,4-Trimethyl pentane

(4) 3-ethyl-2-methyl pentane

Watermarkly

Scanned with OKEN Scanner

NOVA

14)  and relation is :

(1) Diastereomer

(2) Enantiomer

(3) Identical

(4) Structural isomer

15)

The number of total isomers of formula $C_2H_2Br_2$ are :-

(1) 2

(2) 3

(3) 4

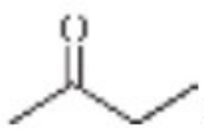
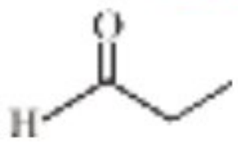
(4) 1

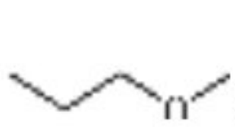
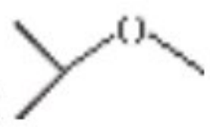
16)

Which is correctly matched ?

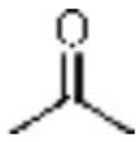
(1)  and  Position isomers

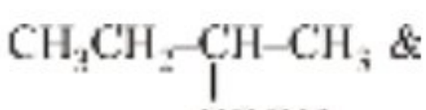
(2)  and 
Functional group isomers

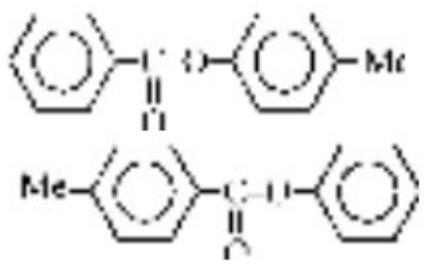
(3)  and  Functional isomers

(4)  and  Chain isomers

17) Identify correct matched ?

(1)  &  - Functional isomers

(2) 
CH_3-CH_2-CH_2-CH_2-COOH - Chain-isomers

(3)  &

Metamers

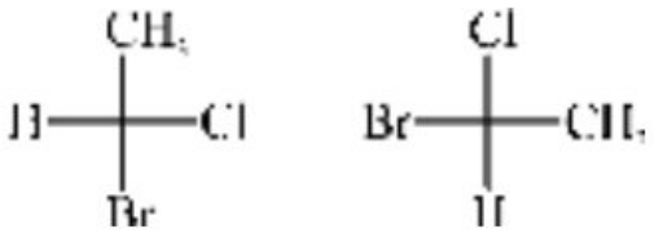
(4) All

18) The following structures represent a pair of :-

@examlix

Watermarkly

Scanned with OKEN Scanner



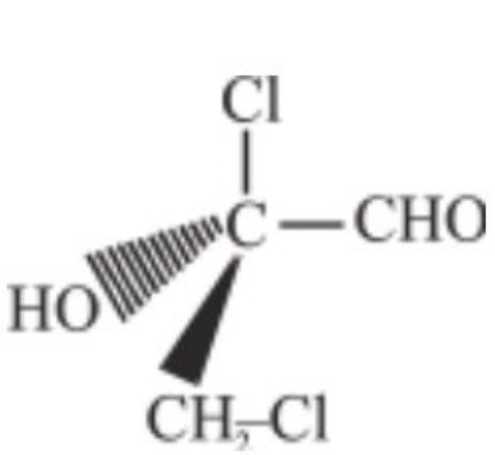
(1) Enantiomer

(2) Diastereomer

(3) Meso compound

(4) Homomers

19) The R-S configuration of following compound will be :-



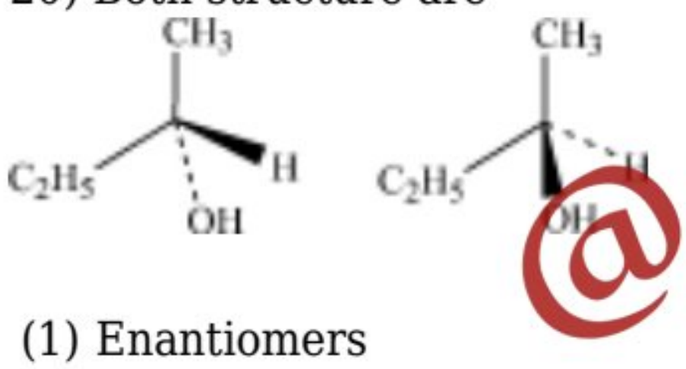
(1) R

(2) S

(3) May R or S

(4) None

20) Both structure are



(1) Enantiomers

(2) Diastereomers

(3) Identical

(4) Position isomers

21) Which form of ethylene glycol will be most stable ?

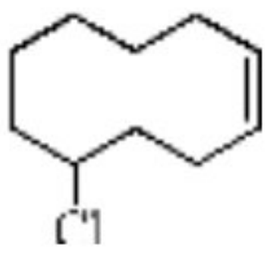
(1) Gauche

(2) Anti form

(3) Fully eclipsed

(4) Partially eclipsed

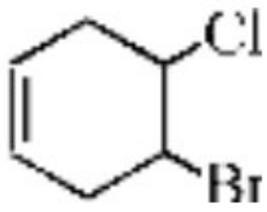
22) Which of the following does not show geometrical isomerism ?

(1) 

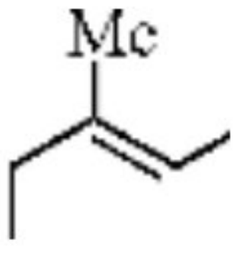
@examlix

Watermarkly

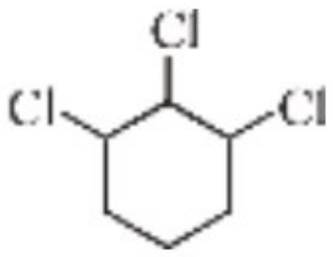
Scanned with OKEN Scanner

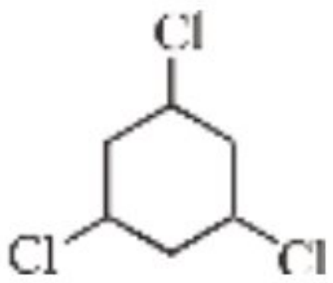
(2) 

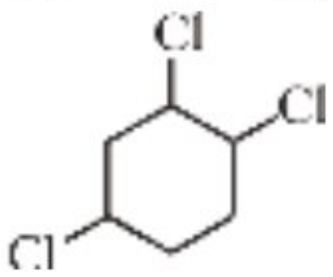
(3) 

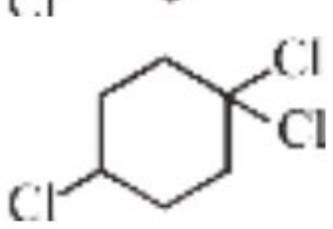
(4) 

23) Which of the following compounds does not have any geometrical isomer ?

(1) 

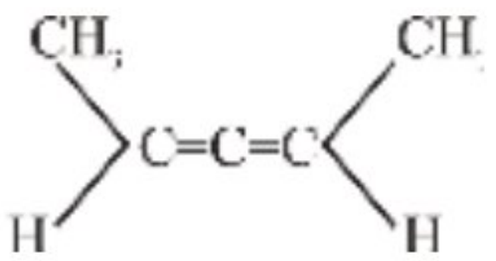
(2) 

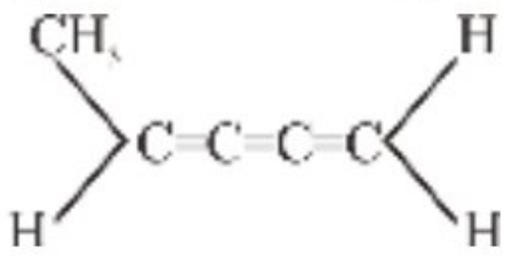
(3) 

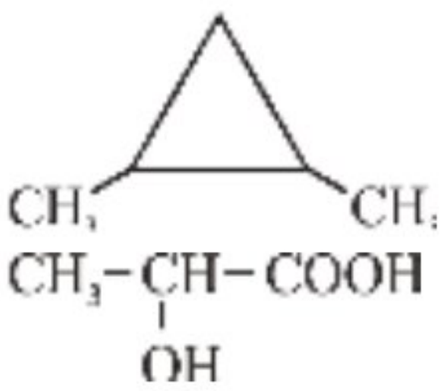
(4) 

@examlix

24) Which can show geometrical and optical isomerism :-

(1) 

(2) 

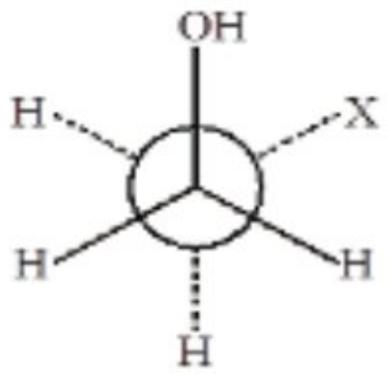
(3) 

(4)

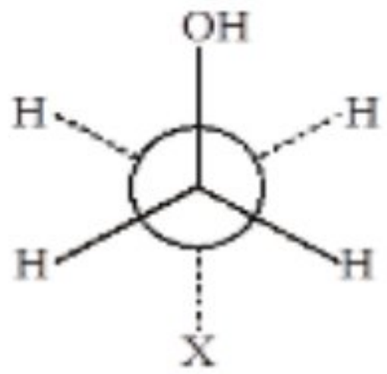
Watermarkly

Scanned with OKEN Scanner

NOVA



(A)



(B)

25)

A is more stable than B, if X is :-

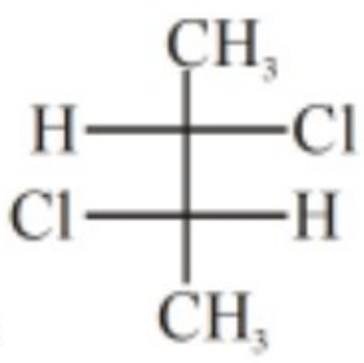
(1) $-CH_3$

(2) F

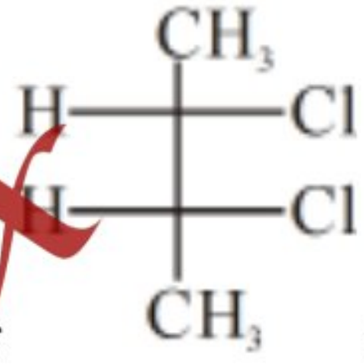
(3) Br

(4) $-C_2H_5$

26)



If angle of rotation of is +52° then angle of rotation of



is :-

(1) 0°

(2) +104°

(3) –52°

(4) Nothing can be said

@examlix

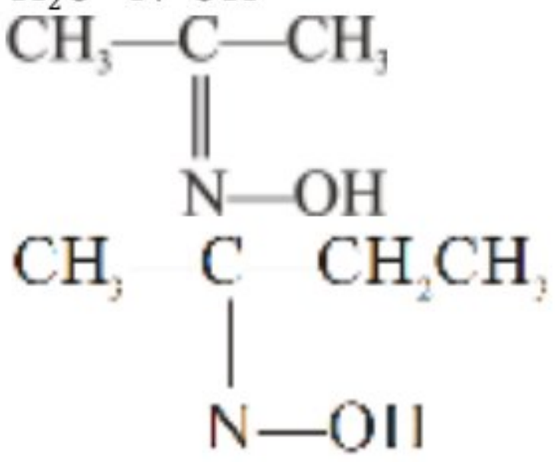
27)

Which of the following shows Geometrical isomerism -

(a) $CH_3–CH_2–CH=N–OH$

(b) $H_2C=N–OH$

(c)



(d)

(1) a, c

(2) c, d

(3) a, d

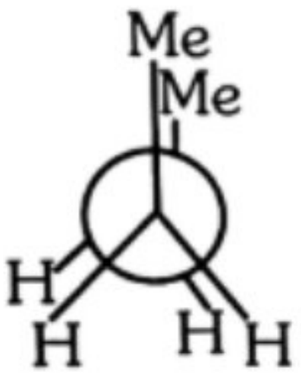
(4) b, c

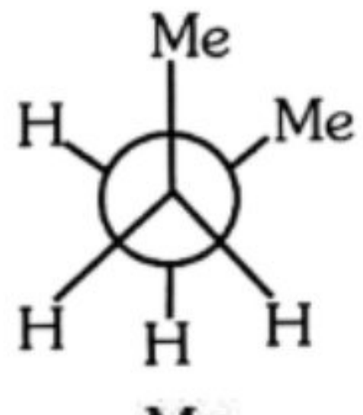
28) Which is the highest energy conformation of butane?

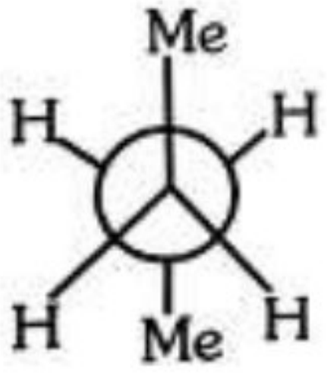
Watermarkly

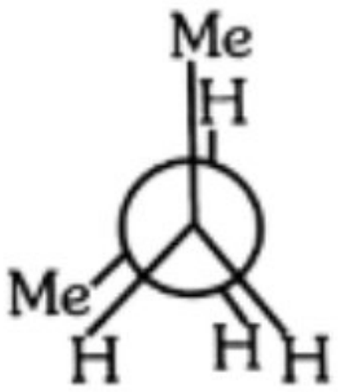
Scanned with OKEN Scanner

NOVA

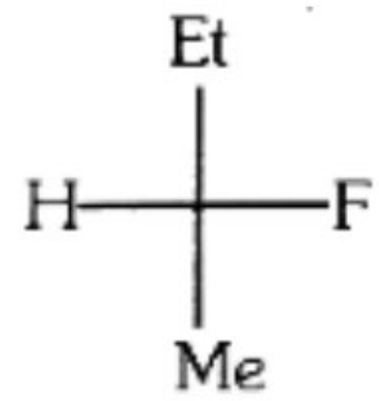
(1) 

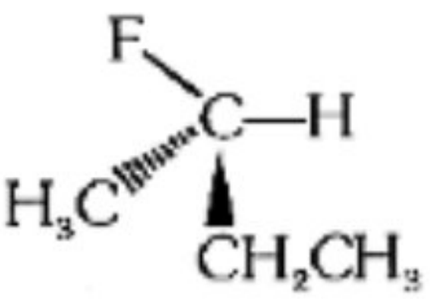
(2) 

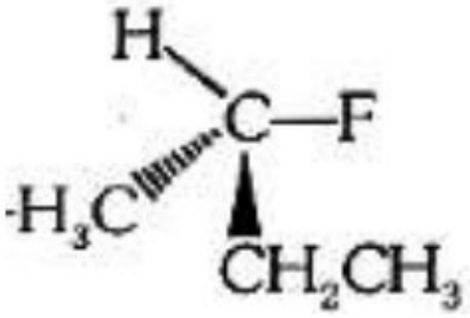
(3) 

(4) 

29) Which of the following has 'S' configuration?

(1) 

(2) 

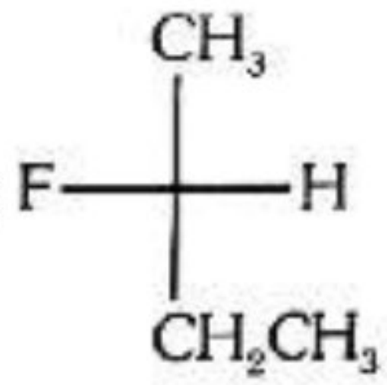
(3) 

@examlix

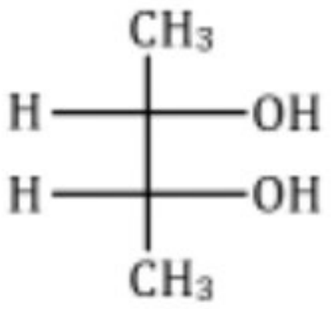
Watermarkly

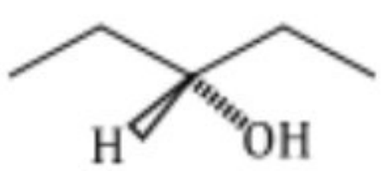
Scanned with OKEN Scanner

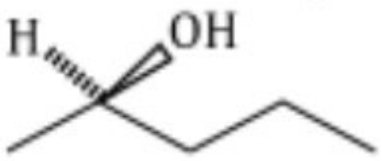
NOVA

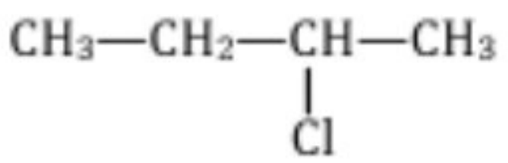
(4) 

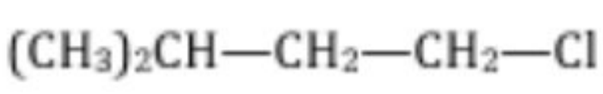
30) Total number of optically active compounds from the following is :-



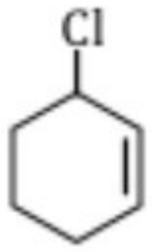












(1) 4

(2) 3

(3) 2

(4) 6

31) Find exothermic reaction :-

(1) $F^-_{(g)} \rightarrow F_{(g)} + \bar{e}$

(2) $Be_{(g)} + \bar{e} \rightarrow B\bar{e}_{(g)}$

(3) $O_{(g)} + \bar{e} \rightarrow O^-_{(g)}$

(4) $O_{(g)} + 2\bar{e} \rightarrow O^{2-}_{(g)}$

32) Correct order of IP will be :-

(1) $F^- > Cl^-$

(2) $F^- < Cl^-$

(3) $O^- > S^-$

(4) Be < B

33) Give the correct initial T for true & F for false statement

(I) For Mn, the order of energy is 3s = 3p = 3d for shell number 3

(II) 32 elements are present in 5th period of long form periodic table

(III) For Zr^{+2} ion, $\mu_m = \sqrt{8}$ BM

(IV) General valence shell electronic configuration for f-block elements is $(n-2)\,f^{1-14}(n-1)\,d^{10}\,ns^2$

(1) T T F T

(2) T F F T

(3) F F T F

@examlix

Watermarkly

Scanned with OKEN Scanner

(4) F T T F

34) Select equation having exothermic step.

(I) $S^{-}_{(g)} \rightarrow S^{-2}_{(g)}$

(II) $Al^{+2}_{(g)} \rightarrow Al^{+3}_{(g)}$

(III) $N(g) \rightarrow N^{-}_{(g)}$

(IV) $Cs^{+}_{(g)} + F^{-}_{(g)} \rightarrow CsF(s)$

Choose the **CORRECT** code.

(1) IV

(2) I, II

(3) III, IV

(4) II, III

35) The correct values of ionization energies (in kJ mol^{-1}) of Be, Ne, He and N respectively are

(1) 899, 2080, 1403, 2372

(2) 2080, 899, 1403, 2372

(3) 899, 2080, 2372, 1403

(4) 899, 1403, 2080, 2372

36) Assuming that Hund's rule is violated, the bond order and magnetic nature of the diatomic molecule B_2 is

(1) 1 and diamagnetic

(2) 0 and diamagnetic

(3) 1 and paramagnetic

(4) 0 and paramagnetic

37) Which of following species has bond order 1.5 ?

(1) SO_4^{-2}

(2) O_2^{-}

(3) CO_3^{-2}

(4) Both (A) & (B)

38) The incorrect hydrated radius order is ?

(1) $Li^{+}(aq) < Be^{2+}(aq)$

(2) $Na^{+}(aq) < Al^{3+}(aq)$

(3) $I^{-}(aq) > Cl^{-}(aq)$

(4) $Ba^{2+}(aq) < Ca^{2+}(aq)$

39) Which is **NOT CORRECT** for covalent character?

@examlix

Watermarkly

Scanned with OKEN Scanner

(1) $SbCl_5 > SbCl_3$

(2) $CaCl_2 > ZnCl_2$

(3) $SnCl_4 > SnCl_2$

(4) $HgCl_2 > ZnCl_2$

40) For BF_3 molecule which of the following is true ?

(1) B-atom is sp^2 hybridised.

(2) There is a pπ–pπ back bonding in this molecule.

(3) Observed B–F bond length is found to be less than the expected bond length.

(4) All of these

41) The orbital diagram in which the Aufbau's principle is violated is

(1) 

(2) 

(3) 

(4) 

@examlix

42) Total number of possible unique sets of four quantum numbers in the 5th shell of an atom, if it is completely filled

(1) 20

(2) 5

(3) 25

(4) 50

43) Which of the following can accommodate maximum number of electron ?

(1) n = 4

(2) All subshells with $n + \ell = 5$

(3) An orbital with n = 3, $\ell = 2$, m = 0

(4) $2p_z$ orbital

44) A p-orbital can accommodate :

(1) 4 electrons

Watermarkly

Scanned with OKEN Scanner

(2) 6 electrons

(3) 2 electrons with parallel spins

(4) 2 electrons with opposite spins

45) Arrange the following orbitals of H-atom in the increasing order of their energy.
$3p_x$, 2s, $4d_{xy}$,3s,$4p_z$,$3p_y$,4s

(1) $2s < 3s = 3p_x = 3p_y < 4s = 4p_z = 4d_{xy}$

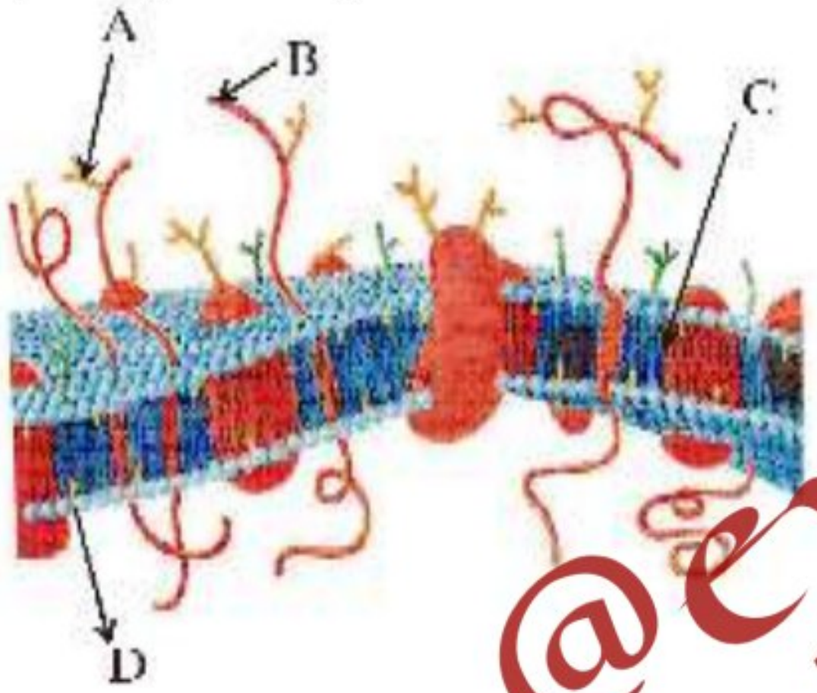
(2) $2s < 3s < 3p_x = 3p_y < 4s = 4p_z = 4d_{xy}$

(3) $2s < 3s < 3p_x = 3p_y < 4s = 4p_z = 4d_{xy}$

(4) $2s < 3s < 3p_x = 3p_y < 4s < 4p_z < 4d_{xy}$

BIOLOGY

1) In a given diagram of cell membrane letter A,B,C and D represents :-



@examlix

	A	B	C	D
(1)	Sugar	Protein	Cholesterol	Lipid bilayer
(2)	Protein	Sugar	Lipid bilayer	Cholesterol
(3)	Sugar	Protein	Lipid bilayer	Cholesterol
(4)	Sugar	Cholesterol	Lipid bilayer	Protein

(1) 1

(2) 2

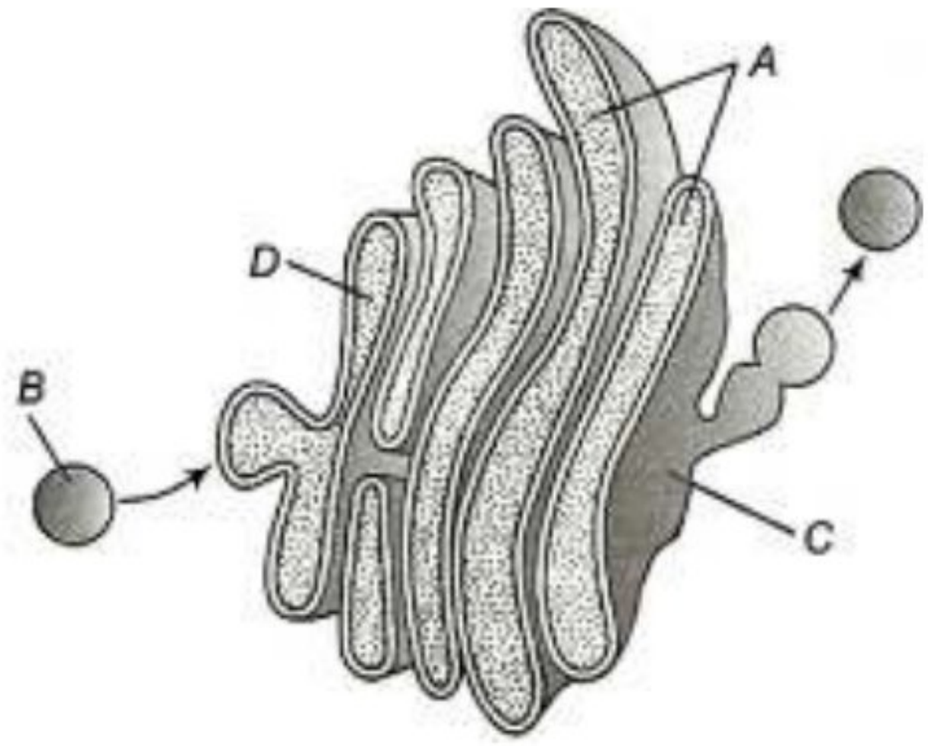
(3) 3

(4) 4

2) Select the option with correct labelling of given structure of Golgi apparatus.

Watermarkly

Scanned with OKEN Scanner



	A	B	C	D
(1)	Cisternae	Vesicle	*trans* face	*cis* face
(2)	Cisternae	Vesicle	*cis* face	*trans* face
(3)	Vesicle	Cisternae	*cis* face	*trans* face
(4)	Tubules	Vesicle	*trans* face	*cis* face

(1) 1

(2) 2

(3) 3

(4) 4

3) Dinoflagellates have -

(1) Only single flagellum, that lies in the transverse groove between the cell plates.

(2) Only single flagellum, that lies in the longitudinal groove between the cell plates.

(3) Two flagella, one lies longitudinally and the other transversely in the furrow between the wall plates.

(4) No flagella.

4) Silica is found in cell wall of :-

(1) Diatoms

(2) Dinoflagellates

(3) Both (1) & (2)

(4) Euglenoids

5) Read the following statements carefully with respect to slime moulds: (A) They produce slimy mass at the time of vegetative phase. (B) Commonly called true fungi.
(C) The spores have true wall.
(D) Plasmodium is formed in unfavourable condition.

(1) A & B are correct

(2) B & C are correct

(3) A & C are correct

@examlix

Watermarkly

Scanned with OKEN Scanner

(4) C & D are correct

6) Which statement is incorrect for fungi?

(1) Reduction division occurs inside fruiting bodies leading to formation of haploid spores
(2) Conidia are produced endogenouly
(3) Formation of ascospores occur inside the ascus
(4) Rust diseases in wheat is caused by *Puccinia*

7) The first discovered antibiotic was obtained from:-

(1) Bacteria
(2) Fungi
(3) Eubacteria
(4) Moneran

8) Which of the following option is incorrect about bacterial pili :-

(1) They are made up of pilin protein
(2) They are shorter than flagella
(3) They help in conjugation
(4) They play an important role in movement

9) The subunit of capsid is called

(1) Core
(2) Nucleotide
(3) Amino acid
(4) Capsomere

10) "Late blight of potato" Disease is caused by:-

(1) A member of Phycomycetes
(2) A member of Ascomycetes
(3) A member of basidiomycetes
(4) A member of deuteromycetes

11) Which of the following is a parasitic fungi on the mustard plant ?

(1) Albugo
(2) Puccinia
(3) Yeast
(4) Ustilago

12) Match the columns I and II, and choose the correct combination from the options given below.

@examlix
Watermarkly
Scanned with OKEN Scanner

	Column I		Column II
(A)	Cyanobacteria	(I)	*Trichoderma*
(B)	Dinoflagellates	(II)	*Albugo*
(C)	Phycomycetes	(III)	*Gonyaulax*
(D)	Deuteromycetes	(IV)	*Anabaena*

(1) A- III, B-I, C-II, D-IV

(2) A- I, B-II, C-III, D-IV

(3) A- IV, B-III, C-II, D-I

(4) A- II, B-IV, C-I, D-III

13) Photosynthetic lamellae of cyanobacteria is called as: -

(1) Chlorophyll

(2) Chromatophore

(3) Chloroplast

(4) Mesosomes

14) Which of the followings is an oxygenic photosynthetic moneran?

(1) *Nostoc*

(2) *E. coli*

(3) *Pseudomonas*

(4) *Rhodospirillum*

@examlix

15) In which of the following process a bacterium acquires genes from an another bacterium by physical contact between them?

(1) Transformation

(2) Transduction

(3) Conjugation

(4) Endospore formation

16) Mad cow disease in cattle is caused by :-

(1) Virus

(2) Viroid

(3) Prions

(4) Virion

17) A location with anormous growth of lichens on the trees indicates that the :-

(1) Trees are very healthy

(2) Trees are heavily infested

(3) Location is highly polluted

Watermarkly

Scanned with OKEN Scanner

(4) Location is not polluted

18) Antherozoids of bryophytes are

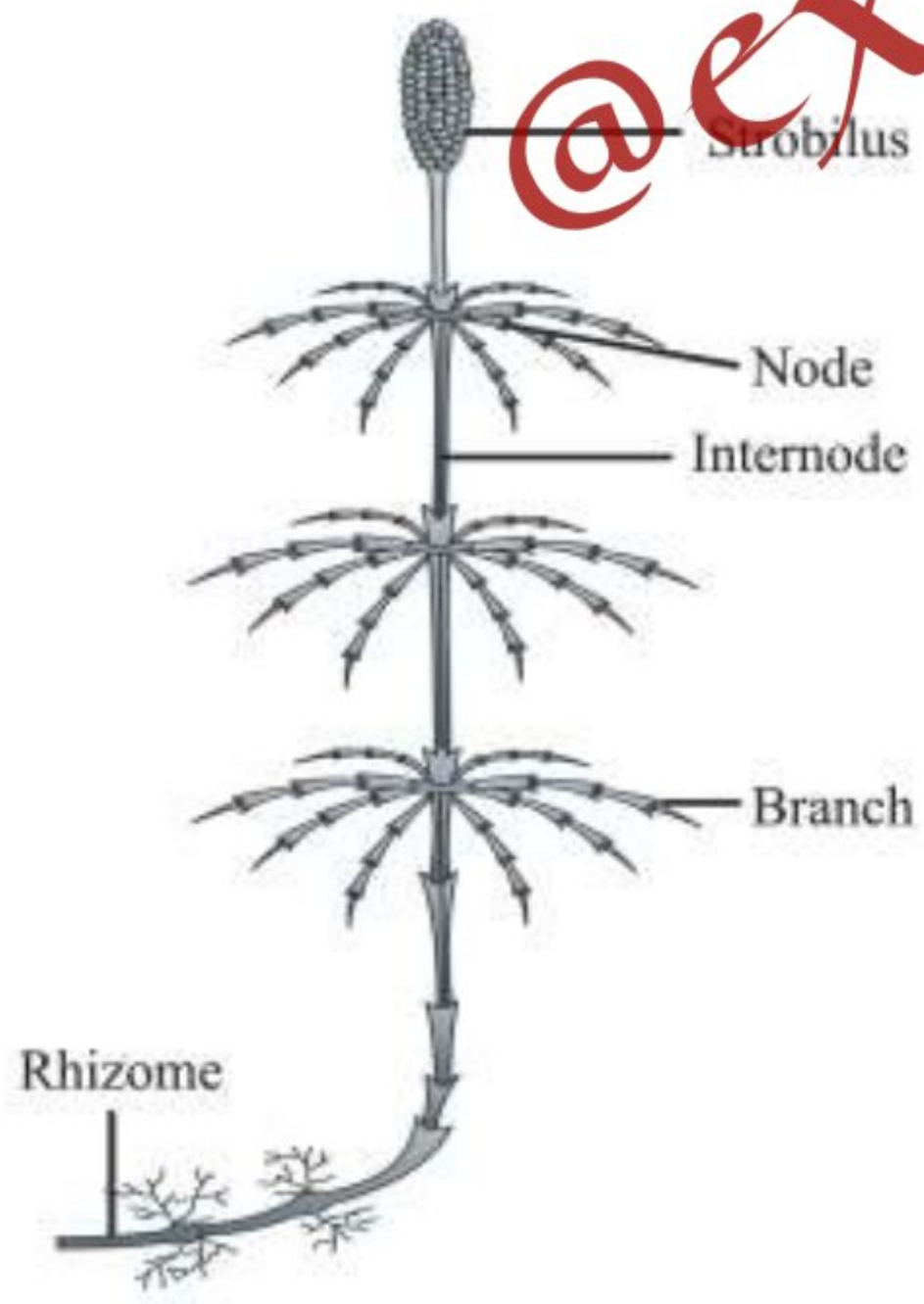
(1) Non-flagellated
(2) Cilliated
(3) Multi cilliated
(4) Biflagellated

19) Match column-I with column-II and select the correct option from the codes given below.

	Column-I		Column-II
A.	Non-vascular plants	(i)	Gymnosperms, Angiosperms
B.	Vascular plants	(ii)	Pteridophytes
C.	Seeded plants	(iii)	Algae, Bryophytes

(1) A-iii, B-ii, C-i
(2) A-ii, B-i, C-iii
(3) A-i, B-ii, C-iii
(4) A-ii, B-iii, C-i

20) Study the given diagram and give answer of questions a, b and c.



(a) Name of plant ? (b) Class of plant ?

(c) Ploidy of rhizome ?

	(a)	(b)	(c)

@examlix
Watermarkly
Scanned with OKEN Scanner

(1)	*Sphagnum*	*Lycopsida*	Haploid
(2)	*Funaria*	Moss	Haploid
(3)	*Funaria*	Liverwort	Diploid
(4)	*Equisetum*	*Sphenopsida*	Diploid

(1) 1

(2) 2

(3) 3

(4) 4

21) *Volvox* and *Fucus* exhibit :-

(1) Oogamy

(2) Isogamy

(3) Anisogamy

(4) Autogamy

22) Coralloid roots of *Cycas* are associated with :-

(1) N_2-fixing cyanobacteria

(2) Stem hair and root cap

(3) Photosynthesis and water conduction

(4) Fungi

23) A protonema is ?

(1) A structure in pteridophytes formed before the thallus develops

(2) A sporophytic free living structure formed in pteridophytes

(3) A creeping, green, filamentous and gametophytic structure produced in bryophytes.

(4) A primitive structure formed after fertilization in pteridophytes

24) One of the most important functions of the plasma membrane is :-

(1) Formation of nuclear membrane

(2) Transport of molecules across it

(3) Exocytosis

(4) Detoxification

25) Rough endoplasmic reticulum differs from smooth endoplasmic reticulum in :-

(1) having ribosomes

(2) Active involvement in protein synthesis

(3) having more cisternae

(4) All of the above

@examlix

Watermarkly

Scanned with OKEN Scanner

26) Different cells have different size. Arrange the following cells in ascending order of their size. Choose correct option among.
(i) Mycoplasma (ii) Ostrech egg
(iii) Human RBC (iv) Bacteria
Options :-

(1) (i), (iv), (iii), (ii)

(2) (i), (ii), (iii), (iv)

(3) (ii), (i), (iii), (iv)

(4) (iii), (ii), (i), (iv)

27) Cells are a structural and functional unit of organisms. Who found out?

(1) Schleiden and schwann

(2) Robert Hook

(3) Aristotle

(4) Mendel

28) Observe the following figures and identify the labelled structures P, Q, R, and S :-



@examlix

	P	Q	R	S
(1)	Archegoniophore	Gemma cup	Antheridiophore	Seta
(2)	Antheridiophores	Rhiziods	Archegoniophore	Capsule
(3)	Antheridiophore	Seta	Archegoniophore	Leaves
(4)	Archegoniophore	Gemma cup	Antheridiophore	Capsule

(1) 1

(2) 2

(3) 3

(4) 4

29) Extensions of cell membrane inside a prokaryotic cell; that are involved in respiration are called ?

(1) Mesosome

(2) Chromatophore

Watermarkly

Scanned with OKEN Scanner

(3) Nucleoid

(4) Plasmid

30) Besides giving out secretory vesicles the golgi apparatus is also concerned with the formation of :

(1) Lysosome

(2) Plastids

(3) Grana of chloroplasts

(4) ER

31) Which of the following character form the basis of modern taxonomic studies?

(1) Structure of cell

(2) Development process of organisms

(3) Ecological information of organisms

(4) All of the above

32) How many are the examples of species in the list given below ?
Angiosperms, Wheat, Rice, Lion, Animals, Dog, Cat, Mango, Reptiles, Amphibians, Man, Diptera, Insecta, Convolvulaceae.

(1) Five

(2) Six

(3) Seven

(4) Four

@examlix

33) Binomial system of nomenclature means that every organism has :-

(1) One scientific nomenclature consisting of two specific epithet

(2) A name given by two scientists

(3) One scientific nomenclature consisting of genera & specific epithet

(4) A number in an international catalogue by which an organism is identified

34) Classification system are changing every now and then because an attempt has been made to evolve such a system ultimately which is able to explain evolutionary relations in organisms. What was the correct sequence of these types of classification systems with respect to their evolution :-

(1) Practical → Artificial → Natural → Phylogenetic → Numerical

(2) Artificial → Practical → Numerical → Phylogenetic

(3) Practical → Artifical → Natural → Numerical → phylogenetic

(4) Numerical → Artifical → Natural → Practical → phylogenetic

35) Which is the common organelles that are found in both eukaryuotes and prokaryote?

(1) Lysosome

(2) Ribosome

Watermarkly

Scanned with OKEN Scanner

(3) Vacuole

(4) Mitochondria

36) Consider the following statements :-

I. All prokaryotic organisms were grouped together under kingdom-Monera.

II. The unicellular eukaryotic organisms were placed in kingdom-Protista.

III. Chlorella and Chlamydomonas, both have cell wall.

IV. Paramoecium and Amoeba lack cell walls.

V. Kingdom-Protista has brought together Chlamydomonas, Chlorella with Paramoecium and Amoeba.

Which of the statements given above are correct?

(1) Only I, II, III and V

(2) Only II, III, IV and V

(3) Only I, II, III and IV

(4) I, II, III, IV and V

37) The structure that helps in attachment to the host tissue in some prokaryotes is :-

(1) Flagella

(2) Fimbrae

(3) Cilia

(4) F- pilli

38) How many kingdoms have only heterotrophic mode of nutrition in Whittaker's classification?

(1) 4

(2) 3

(3) 2

(4) 5

39) In *1969, R.H. Whittaker* proposed five kingdom system of classification. On which basis he classified organisms into five kingdoms i.e. monera, protista, fungi, plantae and animalia ?

(1) Cell structure & body organisation

(2) Body organisation & the mode of nutrition

(3) Reproduction & phylogenetic relationship

(4) All the above

40) **Assertion :** Blue green algae form blooms in polluted water bodies.
Reason : Some of blue green algae can fix atmospheric nitrogen fixation due to presence of heterocysts.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

@examlix

Watermarkly

Scanned with OKEN Scanner

41) **Assertion:**- Boundaries of protista is not well defined
Reason:- What may be photosynthetic protistan to one biologist may be plant to another.

(1) **Assertion** and **Reason** is true and **Reason** is correct explanation of **Assertion**

(2) Only **Assertion** is correct

(3) Only **Reason** is correct

(4) **Assertion** and **Reason** is true but **Reason** is not correct explanation of **Assertion**

42)

	Column I		**Column II**
a.	Methanogens	1.	Salty areas
b.	Halophiles	2.	Marshy areas
c.	Thermoacidophiles	3.	Hot springs

(1) a–2, b–1, c–3

(2) a–1, b–2, c–3

(3) a–3, b–2, c–1

(4) a–2, b–3, c–1

43) The bacterial cell is very simple in ___A___ but very complex in ___B___.
Select the option in which A and B are correctly identified.

	A	**B**
(1)	Behaviour	Structure
(2)	Reproduction	Structure
(3)	Shape	size
(4)	Structure	Behaviour

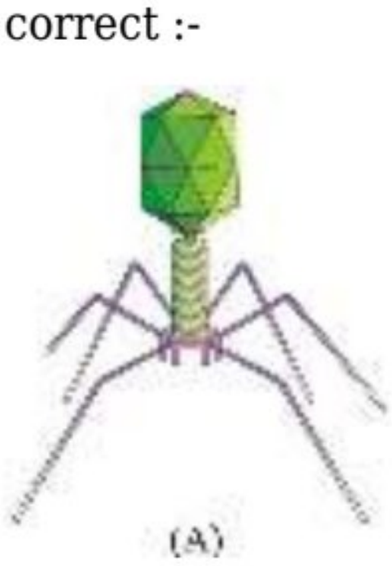
(1) 1

(2) 2

(3) 3

(4) 4

44) Examine the figure A,B,C and D in which one of the four options all the items A,B,C and D are correct :-



(A)



(B)

@examlix

Watermarkly

Scanned with OKEN Scanner



(C)



(D)

Options are :-

	A	B	C	D
(1)	Tobacco mosaic virus	*Aspergillus*	*Agaricus*	Bacterio-phage
(2)	Bacterio-phage	*Agaricus*	*Aspergillus*	Tobacco mosaic virus
(3)	Bacterio-phage	*Aspergillus*	*Agaricus*	Tobacco mosaic virus
(4)	*Agaricus*	Tobacco mosaic virus	Bacterio-phage	*Aspergillus*

(1) 1

(2) 2

(3) 3

(4) 4

@examlix

45) Identify the given figure and select the correct statements regarding it ?



(i) It is *Paramoecium*, having mixotrophic nutrition.
(ii) It is *Euglena* having one functional flagellum.
(iii) It is *Euglena* having pigments identical to higher plants.
(iv) It is *Paramoecium*, a ciliated protozoan.

(1) (i) & (iv)

(2) only (i)

(3) only (iii)

(4) (ii) & (iii)

46) The oxygen-Haemoglobin dissociation curve will show a right shift in case of :-

(1) Less H^+ concentration

(2) High partial pressure of O_2

(3) Low partial pressure of CO_2

(4) High 2,3,D.P.G.

Watermarkly

Scanned with OKEN Scanner

47) Volume of air a person can inspire or expire per minute is -

(1) 2500–3000 ml
(2) 1000–1100 ml
(3) 6000–8000 ml
(4) 1100 – 1200 ml

48) Match the following :-

	Column-A		Column-B
1.	Tidal Volume	A.	TV + IRV + ERV
2.	Residual volume	B.	Additional amount of air inhaled beyond TV during a deep breath
3.	Expiratory reserve volume	C.	Amount of air remaining in lungs after ERV is expelled
4.	Inspiratory reserve volume	D.	TV + IRV
5	Inspiratory capacity	E.	Volume of air in one breath
6	Vital capacity	F.	Amount of air exhaled in forced exhalation

(1) 1-C, 2-E, 3-B, 4-F, 5- D, 6-A
(2) 1-E, 2-F, 3-C, 4-B, 5-A, 6-D
(3) 1-E, 2-C, 3-F, 4-B, 5-D, 6-A
(4) 1-E, 2-C, 3-B, 4-F, 5-A, 6-D

49) CO_2 dissociates from carbamino-haemoglobin when -

(1) pCO_2 is high and pO_2 is low
(2) pO_2 is high and pCO_2 is low
(3) pCO_2 and pO_2 are equal
(4) pCO_2 and pO_2 are high

50) The movement of chloride ions in chloride shift:-

(1) From plasma into RBC
(2) From RBC into plasma
(3) From RBC into W.B.C.
(4) From WBC into plasma

51) What volume of oxygen is transported by one litre of oxygenated blood under normal conditions :-

(1) 150 ml
(2) 50 ml
(3) 200 ml
(4) 100 ml

@examlix
Watermarkly
Scanned with OKEN Scanner

NOVA

52) Which of the following is not a part of respiratory tract

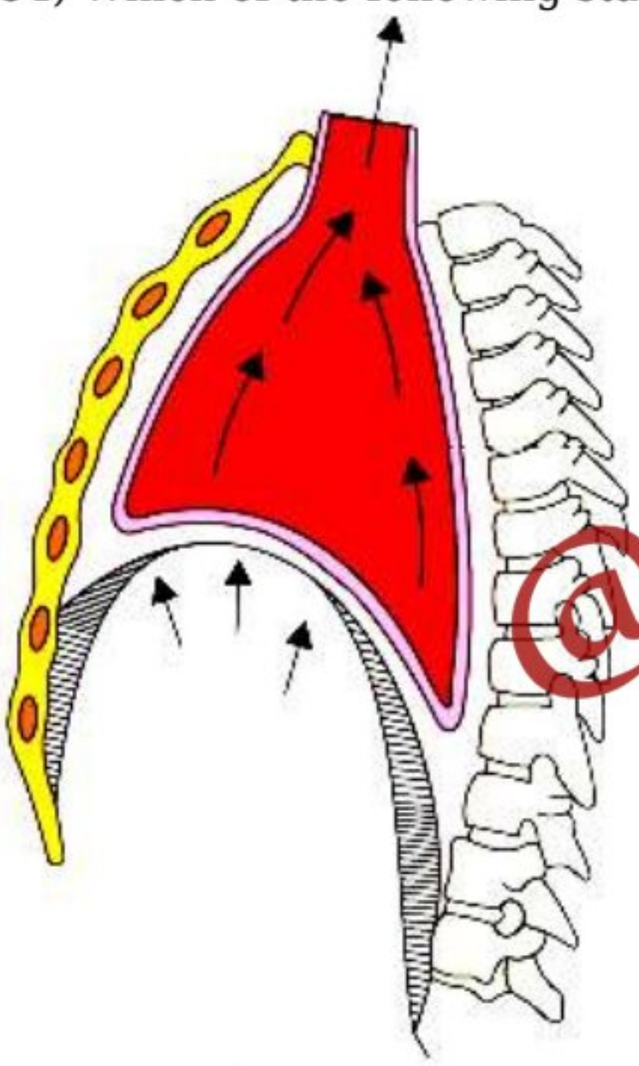
(1) Nasal chamber
(2) Oesophagus
(3) Pharynx
(4) Trachea

53)

Thickness of respiratory membrane is equals to:

(1) 1 mm
(2) Less than 1 mm
(3) More than 1 mm
(4) 2 mm

54) Which of the following statements is **true** regarding the given figure?



@examlix

(1) Diaphragm is contracted
(2) Diaphragm is relaxed
(3) Volume of thorax is increased
(4) Ribs and stemum are raised

55) At which level of vertebral column, does trachea divide in the form of primary bronchi?

(1) 4th thoracic vertebra
(2) 5th thoracic vertebra
(3) 5th cervical vertebra
(4) 7th thoracic vertebra

56) Read the following statements :-

Watermarkly

Scanned with OKEN Scanner

(a) Respiratory rhythm centre is present in Medulla oblongata region.
(b) A chemosensitive area is highly sensitive to CO_2 and hydrogen ions.
(c) The role of oxygen in the regulation of respiratory rhythm is quite insignificant
(d) Pneumotaxic centre is present in Pons varoli region.
(e) Receptors associated with aortic arch and carotid artery can recognise changes in CO_2 and H^+ ion Concentration.
(f) Pneumotaxic centre can reduce the duration of inspiration and thereby alter the respiratory rate.
How many statements are **correct** about regulation of Respiration?

(1) Three
(2) Four
(3) Five
(4) Six

57) In human brain, respiratory centre is activated by :

(1) CO_2 content in venous blood
(2) CO_2 content in arterial blood
(3) O_2 content in arterial blood
(4) O_2 content in venous blood

58) Vascularised bag like structures in lungs are called as ?

(1) Gills
(2) Alveoli
(3) Bronchus
(4) Trachea

59) Pneumotaxic centre is present in

(1) Cerebrum
(2) Cerebellum
(3) Medulla oblongata
(4) Pons varolii

60) Which of the following is **not** the part of bronchial tree ?

(1) Terminal bronchioles
(2) Secondary bronchioles
(3) Tertiary bronchioles
(4) Respiratory bronchioles

61) Find out the amount of O_2 which is delivered by 100 ml of oxygenated blood to the tissue under normal physiological conditions :-

(1) 4 ml
(2) 20 ml

@examlix
Watermarkly
Scanned with OKEN Scanner

NOVA

(3) 5 ml

(4) 25 ml

62) Respiration involves following steps **except** ?

(1) Diffusion across alveolar membrane

(2) Assimilation

(3) Diffusion between blood and tissue

(4) Transport of gases

63) The major fraction of CO_2 released during cellular respiration is transported by the blood to the lungs.

(1) in combination with Hb

(2) As free CO_2

(3) As carbonic acid

(4) In the form of bicarbonate ions

64) What amount of carbon dioxide is delivered by every 100 ml of deoxygenated blood to the alveoli?

(1) 4 ml

(2) 5 ml

(3) 15 ml

(4) 20 ml

@examlix

65) Partial pressure of oxygen in alveoli and deoxygenated blood is respectively (in mm Hg)

(1) 159 and 104

(2) 104 and 40

(3) 95 and 40

(4) 40 and 45

66) Oxygen dissociation curve shift to left in which condition ?

(1) High P_{CO_2}

(2) High P_{O_2}

(3) Low H^+ concentration

(4) Both (2) and (3)

67) During breathing, inspiration can occur :-

(1) If intra-pulmonary pressure is less than the atmospheric pressure

(2) If intra-pulmonary pressure is more than the atmospheric pressure

Watermarkly

Scanned with OKEN Scanner

(3) If there is a negative pressure in the lungs with respect to atmospheric pressure
(4) (1) and (3) both

68) Which is a chronic disorder about respiration :-

(1) Asthma
(2) Emphysema
(3) Fibrosis
(4) Silicosis

69) The lungs are made up of :-
(a) Bronchi (b) Bronchiales
(c) Alveoli (d) Trachea

(1) a and b
(2) a, b and c
(3) b, c and d
(4) a, b and d

70) *Pavo* (Peacock) is a bird that belongs to class Aves. It has feathers that help in protective colouration and keeping the body warm. Pneumatic, long bones are present that helps it in flying. Which of the following statement is incorrect w.r.t *Pavo* is incorrect?

(1) Heart is ventral with four chambers.
(2) Notochord is persistent throughout the life.
(3) Crop and gizzard are additional chambers present.
(4) They are homoiothermic in nature.

71) How many of the following statements are false?
a. Gills are without operculum in Chondrichthyes.
b. Non-chordates possess dorsal, solid and double nerve cord.
c. Heart in non-chordates is dorsal and always present.
d. Water vascular system helps in reproduction.
e. Alimentary canal, urinary and reproductive tracts open into cloaca in amphibians.

(1) Four
(2) Two
(3) Three
(4) Five

72) Read the following statements and choose the correct option:
Assertion (A) : Chordates are bilaterally symmetrical.
Reason (R) : The body can be divided into identical left and right halves in only one plane.

(1) Both assertion and reason are true, and reason is the correct explanation of assertion.
(2) Both assertion and reason are true, but reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.

@examlix
Watermarkly
Scanned with OKEN Scanner

(4) Both assertion and reason are false.

73) Choose the correct option –
Assertion (A) : Mammals commonly exhibit viviparity.
Reason (R) : Mammary glands are milk producing glands by which the young ones are nourished.

(1) Both assertion and reason are true, and reason is the correct explanation of assertion.
(2) Both assertion and reason are true, but reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Both assertion and reason are false.

74) Identify the given figure and mark the statement which is incorrect:



(1) They are oviparous organisms.
(2) Lungs helps in respiration.
(3) They possess mammary glands for nourishment of their young ones.
(4) Heart is three chambered.

@examlix

75) One of the representatives of Phylum Arthropoda is :

(1) flying fish
(2) cuttlefish
(3) silverfish
(4) pufferfish

76) **Assertion :** Tapeworms absorb digested food by its body surface.
Reason : Tapeworms possess digestive organs.

(1) Both assertion and reason are true and the reason is a correct explanation of the assertion.
(2) Both assertion and reason are true and the reason is not a correct explanation of the assertion.
(3) Assertion is true but the reason is false.
(4) Assertion is false but reason is true.

77) Read the following statements w.r.t. Echinodermata :
(i) Are exclusively marine
(ii) Digestive system in incomplete
(iii) Tube feast is universally absent
(iv) Are spiny bodied
(v) The adults are radially symmetrical but larva are bilaterally symmetrical
Which of the above statements are correct?

Watermarkly

Scanned with OKEN Scanner

(1) (i), (iv), (v)
(2) (i), (ii), (iii)
(3) (ii), (iii)
(4) (iv) and (v) only

78) Which of the following is not a mollusc?

(1) Chiton
(2) Sea-hare
(3) Devil fish
(4) Sea-fan

79) Which of the following are sensory organs in phylum Arthropoda?
(a) Simple or compound eye
(b) Statocyst
(c) Malpighian tubules
(d) Antennae
(e) Anal cerci

(1) All except (b)
(2) All except (e)
(3) All except (c)
(4) All of the above

80) Choose the incorrect statements from the following:
I. Function of GLUT-4 is to enable glucose transport into the cells.
II. Neural tissues have lipids with more complex structure.
III. Lipids are strictly macromolecules.
IV. Metabolic pathways of a living system crisscross each other.
V. Bioenergetics deals with how and in which direction the metabolites flows in a metabolic pathway.

(1) II, III, IV and V only
(2) IV and V only
(3) I, III, IV and V only
(4) III and V only

81) Following are the statements with reference to 'lipids'.
(a) Lipids having only single bonds are called unsaturated fatty acids.
(b) Lecithin is a phospholipid.
(c) Trihydroxy propane is glycerol.
(d) Palmitic acid has 20 carbon atoms including carboxyl carbon.
(e) Arachidonic acid has 16 carbon atoms.
Choose the correct answer from the options given below.

(1) (b) and (c) only
(2) (b) and (e) only

@examlix
Watermarkly
Scanned with OKEN Scanner

(3) (a) and (b) only

(4) (c) and (d) only

82) **Statement 1:** protein imagined as a line, left end represent first amino acid known as N terminal.
Statement 2: right end represent last amino acid known as C terminal

(1) Both statement 1 and statement 2 are correct

(2) Statement 1 is correct and statement 2 is incorrect

(3) Statement 1 is incorrect and statement is correct

(4) Both statements are wrong

83) Given below are two statements:
Statement I : Amino acids are called alpha amino acids as they contain an amino group and an acidic group on the same carbon.
Statement II: A proteinaceous amino acid which contain a methyl group is serine. In the light of the above statements,
choose the most appropriate answer from the options given below :

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct.

84) Following are the statements with reference to 'lipids'.
(a) Lipids having only single bonds are called unsaturated fatty acids.
(b) Lecithin is a phospholipid.
(c) Trihydroxy propane is glycerol.
(d) Palmitic acid has 20 carbon atoms including carboxyl carbon.
(e) Arachidonic acid have only single bond.
Choose the correct option which contains **correct** statements.

(1) (a) and (b) only

(2) (c) and (d) only

(3) (b) and (c) only

(4) (b) and (e) only

85) Read the following statements (A-D).
(A) DNA is an acidic substance
(B) The two chains of DNA have antiparallel polarity
(C) 5-Methyl uracil is another chemical name of thymine
(D) The two chains of B-DNA are coiled in a right handed fashion
How many of above statements are correct?

(1) 4

(2) 3

(3) 2

@examlix

Watermarkly

Scanned with OKEN Scanner

(4) 1

86) Identify the incorrect statement regarding the structure of proteins:

(1) They are heteropolymers containing strings of amino acids
(2) The first amino acid in the primary structure is the N-terminus amino acid
(3) Only left helices are observed in proteins
(4) Tertiary structure is absolutely essential for many biological functions of the proteins

87) With the increase in substrate concentration, the velocity of enzyme action

(1) Rises at first and then slows down
(2) Rises at first and reaches a maximum velocity
(3) Lowers at first and reaches a maximum velocity
(4) Lowers at first and then rises

88) Match the following columns

	Column I (Components)		**Column II (% of the totalcellular mass)**
A	Water	I	10- 15
B	Proteins	II	70-90
C	Carbohydrates	III	5 -7
D	Lipids	IV	3
E	Nucleic acids	V	2

	A	**B**	**C**	**D**	**E**
1	I	III	II	IV	V
2	II	I	IV	V	III
3	I	II	III	IV	V
4	V	III	II	I	IV

(1) 1
(2) 2
(3) 3
(4) 4

89) Represented below is the effect of change in pH on enzyme activity. Which one of the following

@examlix

Watermarkly

Scanned with OKEN Scanner



conclusions can be made by this diagram ?

(1) Enzyme show its highest activity at a particular pH, both below and above this pH, activity declines

(2) With increasing pH, activity increases upto a limit then it become constant

(3) At a particular pH, activity is minimum, both below and above this pH activity increases

(4) Enzyme activity is affected by temperature not by pH so it is completely a hypothetical graph

90) According to enzyme substrate complex theory :-

$$\text{Enzyme [E]} + \text{Substrate [S]} \rightleftharpoons A \rightarrow B \rightarrow E + P \quad [P = \text{Product}]$$

In the given reaction A and B is :-

(1) A → EP complex, B → ES complex

(2) A → ES complex, B → EP complex

(3) A → EP complex, B → EP complex

(4) A → ES complex, B → ES complex

@examlix

Watermarkly

Scanned with OKEN Scanner

NOVA

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	4	1	1	3	2	3	4	3	3	1	4	2	2	4	3	2	1	1	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	3	4	1	1	3	4	3	1	3	3	3	1	3	2	1	4	2	3	2
Q.	41	42	43	44	45															
A.	2	2	1	1	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	1	1	4	2	3	3	1	3	1	2	2	1	4	2	1	4	4	2	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	3	4	3	2	1	3	1	3	2	3	2	3	1	3	1	4	3	2	4
Q.	86	87	88	89	90															
A.	2	4	1	4	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	1	3	1	3	2	2	4	4	1	1	3	2	1	3	3	4	4	1	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	1	3	2	4	1	1	4	1	1	4	3	3	1	2	4	2	3	4	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	4	2	4	4	3	3	2	1	3	2	2	2	2	4	2	2	4	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	4	1	2	4	4	2	2	2	3	1	2	4	3	3	1	4	3	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	1	2	3	1	3	2	2	1	2										

@Examlix

Watermarkly

Scanned with OKEN Scanner

SOLUTIONS

PHYSICS

1) $P_{mix} = P_1 + P_2$ $P_{mix} = \dfrac{\mu_1 RT_1}{V_1} + \dfrac{\mu_2 RT_2}{V_2}$

$= \dfrac{1 \times 25 \times 300}{3 \times 5 \times 10^{-3}} + \dfrac{1 \times 25 \times 300}{3 \times 5 \times 10^{-3}}$

$= 10 \times 10^5\ N/m^2$

$P_{mix} = 10$ atm

2) **Question Explanation**
In this question, we are given a equation for a thermodynamic process and asked to find the molar specific heat of a diatomic gas during this process.

Concept
Molar specific heat of gas in Polytropic process.

Solution
Given:
$P^3V^2 = \text{constant} \Rightarrow PV^{2/3} = \text{constant}$

So, for a polytropic process
$PV^n = \text{constant}$

Comparing with above equation, we get

$\Rightarrow n = \dfrac{2}{3}$

For a polytropic process, molar specific heat is given by

$C = \dfrac{f}{2}R + \dfrac{R}{1-n}$

For a diatomic gas , f = 5

$C = \dfrac{5}{2}R + \dfrac{R}{1-\left(\frac{2}{3}\right)}$

$C = \dfrac{5}{2}R + \dfrac{3R}{1}$

$C = \dfrac{5R + 6R}{2} = \dfrac{11R}{2}$

Final Answer: Option (4)

3) **Explain:** find the temperature at the common wall between two walls of different thicknesses and thermal conductivities, given the temperatures of the outer surfaces.
Concept: In steady state, the rate of heat flow through both walls must be equal.
Formula: Rate of heat flow (conduction): $\dfrac{Q}{t} = kA\dfrac{\Delta T}{d}$

@examlix
Watermarkly
Scanned with OKEN Scanner

$\frac{Q}{t}$

Where $\frac{Q}{t}$ is the rate of heat flow

k is the thermal conductivity

A is the cross-sectional area

ΔT is he temperature difference across the wall

Calculation: In steady state, $\left(\frac{Q}{t}\right)_1 = \left(\frac{Q}{t}\right)_2$

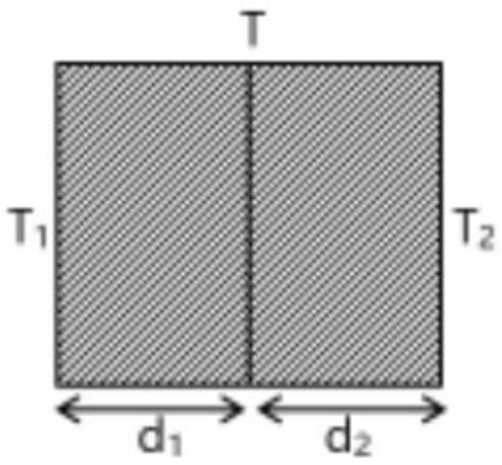
$$\frac{k_1 A (T_1 - - T)}{d_1} = \frac{k_2 A (T - - T_2)}{d_2}$$

$k_1(T_1 - T)d_2 = k_2(T - T_2)d_1$

$$T = \frac{(k_1 T_1 d_2 + k_2 T_2 d_1)}{(k_1 d_2 + k_2 d_1)}$$

Ans. (1)

Difficulty level: moderate



4)

Correct Answer (1)

@examlix

5) **Question Explanation:** In the provided P-V diagram, four different thermodynamic processes (I, II, III, and IV) originate from an initial state i and move toward final states f across isotherms of 300 K, 500 K, and 700 K. Match each process curve to its corresponding thermodynamic type: Adiabatic, Isobaric, Isochoric, or Isothermal.

Concept: Thermodynamic Processes on a P-V Diagram

Formula:

Isobaric: P = constant

Isochoric: V = constant

Isothermal: PV = constant (Path follows a single temperature curve)

Adiabatic: PV^{γ} = constant (Steeper slope than isothermal)

Solution:

Process I (P→c): The line is vertical, meaning the volume (V) does not change as pressure drops. This is an Isochoric process.

Process II (Q→a): This curve drops more sharply than the isothermal curve (Process III). In thermodynamics, the slope of an adiabatic curve is γ times steeper than an isothermal one ($\left|\frac{dP}{dV}\right|_{adia} = \gamma \left|\frac{dP}{dV}\right|_{iso}$). Therefore, this is an Adiabatic process.

Process III (R→d): This curve follows the path from initial state i (on the 500 K isotherm) to a final state f that stays on the same 500 K isotherm. Since temperature remains constant, this is an Isothermal process.

Watermarkly

Scanned with OKEN Scanner

Process IV (S→b): The line is horizontal, meaning the pressure (P) remains constant as volume increases. This is an Isobaric process.
Final answer: Option (3)

6) **Explain Question:**
In this question, we are asked the final temperature when ice at 0°C is mixed with hot water.

Concept:
Principle of Calorimetry- Mixing of different phases

Solution:
Given:
• Ice mass = 50 g at 0°C
• Water mass = 50 g at 60°C
• S_w = 4.2 J/g°C
• L_f = 334 J/g
Heat released by water (cooling to 0°C):
$Q_w = ms\Delta T = 50 \times 4.2 \times 60 = 12600$ J
Heat required to melt entire ice:
$Q_i = mL_f = 50 \times 334 = 16700$ J
Since
$Q_w < Q_i$
water does not have enough heat to melt all the ice, so mixture stays at 0°C.

Final Answer: Option (2)

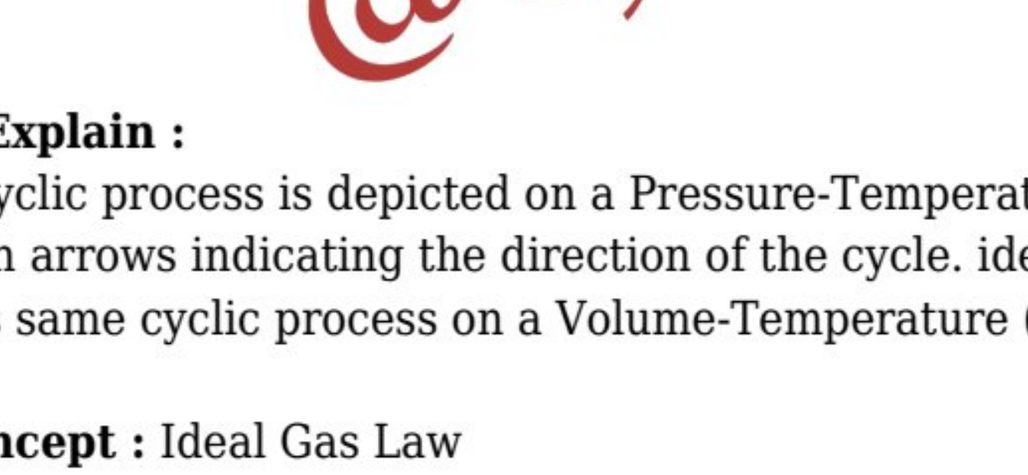
@examlix

7) **Explain :**
A cyclic process is depicted on a Pressure-Temperature (P-T) diagram as a rectangle ABCD with arrows indicating the direction of the cycle. identify the corresponding representation of this same cyclic process on a Volume-Temperature (V-T) diagram, assuming an ideal gas.

Concept : Ideal Gas Law

Formula : PV = nRT

Calculation :



A. **A → B (Constant P, T increases):** $V \propto T$ (straight).

B. **B → C (Constant T, P decreases):** $V \propto \frac{1}{P}$. V is inversely proportional to P. As P decreases, V

Watermarkly

Scanned with OKEN Scanner

increases (VERTICLE line).

C. **C → D (Constant P, T decreases):** $V \propto T$ (straight line).

D. **D → A (Constant T, P increases):** $V \propto \frac{1}{P}$. V is inversely proportional to P. As P increases, V decreases (horizontal line to the left on V-T).

Correct Answer :
Option (3)

8) **Explain :**
IDENTIFY the correct expression(s) for the ratio of specific heats (γ) of an ideal gas in terms of its specific heat at constant pressure (C_p), specific heat at constant volume (C_v), and the ideal gas constant (R).

Concept : Specific heats of an ideal gas and Mayer's relation.

Formula :

$$\gamma = \frac{C_p}{C_v}$$

$$C_p - C_v = R$$

Calculation:
Option 1 :

$$\frac{1}{1-\left(\frac{R}{C_p}\right)} = \frac{1}{\frac{C_p-R}{C_p}} = \frac{C_p}{C_p-R} = \frac{C_p}{C_v} = \gamma$$

Option 2 :

$$1 + \frac{R}{C_v} = \frac{C_v+R}{C_v} = \frac{C_p}{C_v} = \gamma$$

Option 3:

$$\frac{C_p}{(C_p-R)} = \frac{C_p}{C_v} = \gamma$$

Final Answer:
Option (4)

9) $\frac{T_1}{T_2} = \frac{\lambda_2}{\lambda_1}$

$$\frac{T_1}{T_2} = \frac{4800}{3600} = \frac{4}{3}$$

10)

$P \propto T$

$$\frac{\Delta P}{P} \times 100 = \frac{\Delta T}{T} \times 100 = 0.4 = \frac{1 \times 100}{T}$$

$$T = \frac{1000}{4} = 250\,K$$

@examlix
Watermarkly
Scanned with OKEN Scanner

11)

Question Explanation
Asking about: The time required for a body to undergo a second temperature drop (70ºC to 40ºC), given its initial cooling performance (100ºC to 70ºC in 4 minutes) and a constant room temperature of 15ºC.

Concept
Newton's Law of Cooling (Average Form)

$$\frac{T_1 - T_2}{t} = K\left[\left(\frac{T_1 + T_2}{2}\right) - T_s\right]$$

Solution
First cooling ($100 \rightarrow 70$ in 4 min):

$$\frac{100-70}{4} = K\left[\frac{100+70}{2} - 15\right]$$

$$\frac{30}{4} = K(85-15)$$

$$\frac{30}{4} = K(70)$$

Second cooling ($70 \rightarrow 40$ in t):

$$\frac{70-40}{t} = K\left[\frac{70+40}{2} - 15\right]$$

$$\frac{30}{t} = K(55-15)$$

$$\frac{30}{t} = K(40)$$

Divide equations:

$$\frac{30/t}{30/4} = \frac{40}{70}$$

$$\frac{4}{t} = \frac{40}{70} \Rightarrow t = \frac{4(70)}{40} = 7 \text{ min}$$

Final Answer: Option (1)

12) **Question Explanation:**
We need the temperature at which oxygen molecules have the same rms speed as hydrogen molecules at 200 K.

Concept:

Root mean square speed: $v_{rms} = \sqrt{\frac{3RT}{M}}$.

Solution:
Given $M_{O_2} = 32$ g/mol, $M_{H_2} = 2$g/mol.
Set v_{rms} equal:

@examlix
Watermarkly
Scanned with OKEN Scanner

$$\sqrt{\frac{3RT_{O_2}}{M_{O_2}}} = \sqrt{\frac{3RT_{H_2}}{M_{H_2}}}$$

Cancel 3R:

$$\frac{T_{O_2}}{M_{O_2}} = \frac{T_{H_2}}{M_{H_2}}$$

$$\frac{T_{O_2}}{32} = \frac{200}{2} = 100$$

$$T_{O_2} = 3200\ K$$

Final Answer: Option (4)

13) $\frac{d\theta}{dl} = 80$

$$\int d\theta = \int 80 dl$$

$30 - \theta = 80 \times 0.5$

$\theta = -10\ °C$

14) **Question Explanation :** Identify the specific mechanism of heat transfer—among conduction, convection, and radiation—that relies on the physical movement or actual mass motion of molecules from a region of higher temperature to a lower temperature.

Concept: Heat Transfer Mechanisms

Solution:

Convection is the mode of heat transfer that occurs primarily in fluids (liquids and gases). It involves the actual physical movement or "mass motion" of the heated molecules themselves. As fluid is heated, it becomes less dense and rises, while cooler, denser fluid moves in to take its place, creating a circulation pattern.

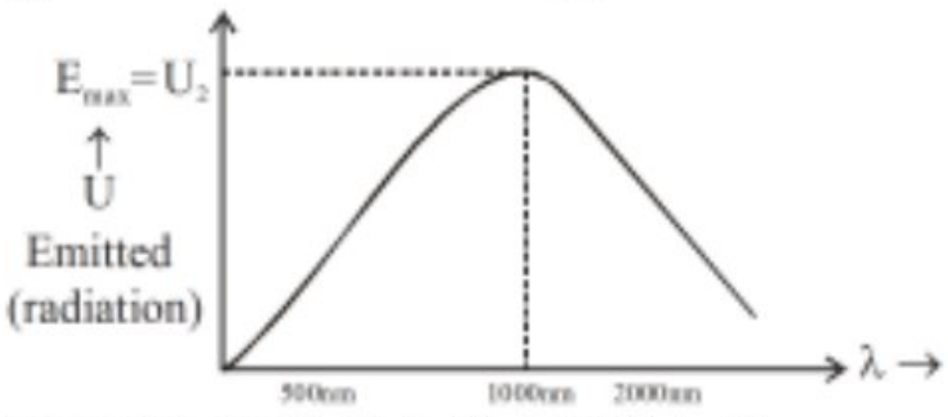
Conduction involves the transfer of thermal energy through direct contact between particles without the overall movement of the material itself.

Radiation involves the transfer of energy through electromagnetic waves, which requires no medium and involves no motion of matter to carry the heat.

Final answer: Option (2)

15) Maximum amount of emitted radiation corresponding to $\lambda_m = \frac{b}{T}$

$$\lambda_m = \frac{2.88 \times 10^6 nmK}{2880K} = 1000\ nm$$



From the graph $U_2 > U_1$ and $U_2 > U_3$

16)

Watermarkly

Scanned with OKEN Scanner

Question Explanation: Check how molar heat capacity behaves for different thermodynamic processes.

Concept:

$$C = \frac{dQ}{n\,dT}$$

Solution:

(a) Adiabatic process

$$dQ = 0 \Rightarrow C = 0$$

Correct.

(b) Isothermal process

$$dT = 0, dQ \neq 0 \Rightarrow C = \infty$$

Correct.

(c) Constant volume or pressure,

$$C_v = \frac{f}{2}R, C_p = \left(\frac{f}{2}+1\right)R$$

C_V, C_P depend on nature of gas

Correct.

(d) Relation

$$C_{molar} = (\text{molecular weight}) \times (\text{gram specific heat})$$

Valid definition of molar heat capacity. Correct.

Final Answer: Option(3)

17) $$\Delta U = \frac{nfR\Delta T}{2} = 0$$

$\Rightarrow \Delta T = 0$

$T_f = T_i$

$P_f V_f = P_i V_i$

18)

Question Explanation : Asking about the correct graph showing the isothermal expiation then isobaric compression.

Concept : This question is based on indicator diagram PV graph.

Solution :

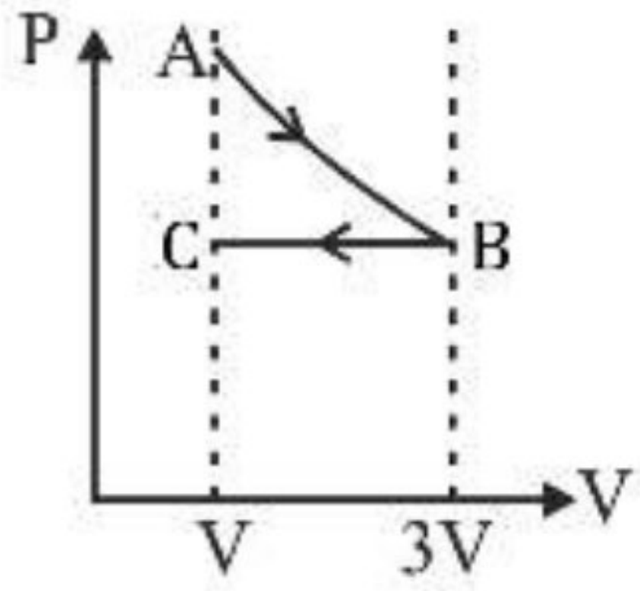
As Given in the question:

Process AB will an isothermal expansion $\left(P \propto \frac{1}{V}\right)$ from V to 3V → rectangular hyperbola

@examlix
Watermarkly
Scanned with OKEN Scanner

curve.
Process BC will an isothermal an isobaric (P = Constant) compression from 3V to V →
Horizontal straight line curve

Then correct graph is :



Final Answer: option (1)

19)

Explaining the Question :- Water is heated on the top of Mount Everest where atmospheric pressure is very low. We have to determine the temperature at which water will start boiling.
Concept:- This question is based on Boiling point of a liquid decreases when external pressure decreases.
Solution :- Boiling occurs when vapour pressure of water equals atmospheric pressure.
On Mount Everest, atmospheric pressure is much lower that at sea level, so this condition is achieved at a lower temperature than 100°C. Hence, water starts boiling below 100°C.
Final Answer :- 1

20)

Correct Answer is Option (1).

21)

Correct Answer is Option (1).

22) No spontaneous heat transfer from a cold body to a hot body is possible.

23) **Concept Based:**
Force on a Pulley (Tension and Gravity Effect)
The pulley experiences two forces:
• The tension force T due to the hanging mass.
• The combined force mg + T due to the pulley's own weight and the tension acting vertically downward.
The force applied by the clamp is the resultant of these forces.

Examlix
Watermarkly
Scanned with OKEN Scanner

Formula Used:
The net force exerted by the clamp is given by:

$$F = \sqrt{T^2 + (mg + T)^2}$$

Calculation:
• Given that T = Mg, the two forces acting on the pulley are:
1. Horizontal force: T = Mg
2. Vertical force: mg + T = mg + Mg = (M + m)g
• The resultant force exerted by the clamp:

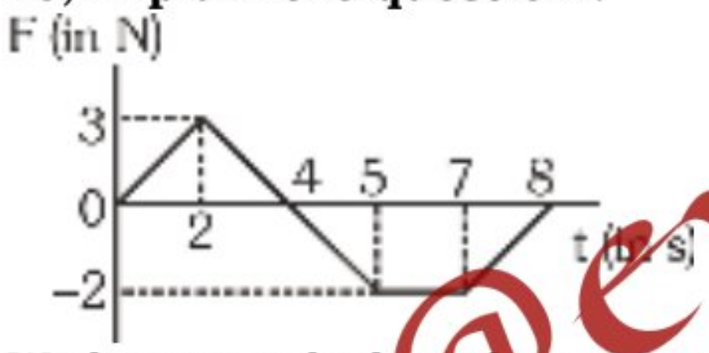
$$F = \sqrt{T^2 + (mg + T)^2}$$

$$F = g\sqrt{M^2 + (M + m)^2}$$

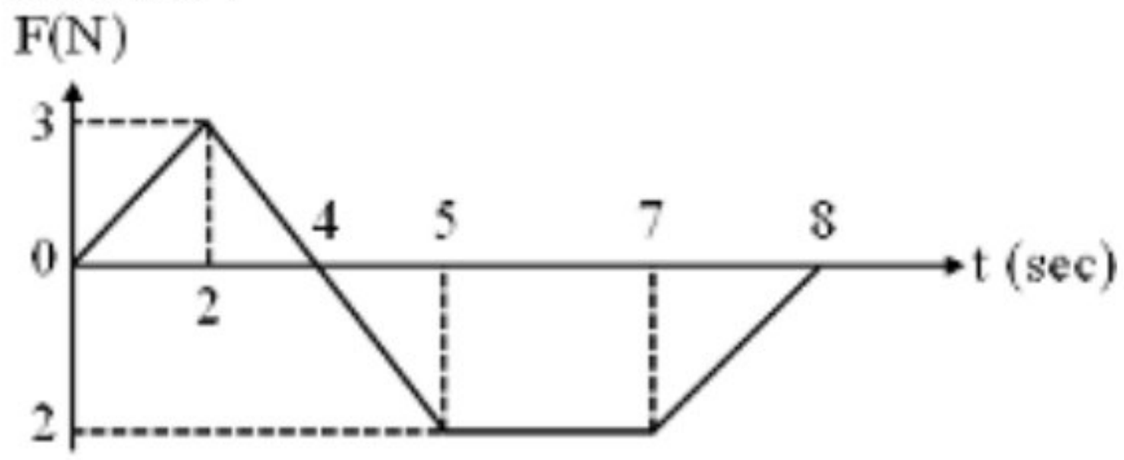
Final Answer: The correct option is (4).

24)

25) **Explain the question :**



We have to calculate change in momentum between t = 0 to t = 9 sec.
Concept : This question is based on area under the curve of force-time graph is equal to change in momentum.
Solution :



ΔP = area between t = 0 to t = 8 sec
$\Delta P = |A_1| - |A_2|$

$$\Delta P = \frac{1}{2} \times 4 \times 3 - \frac{1}{2}[4 + 2] \times 2$$

$\Delta P = 6 - 6 = 0$
Final answer : 1

26)

Question Explanation :
Calculate the initial acceleration of a 60 kg rocket in gravity-free space as it ejects gas at a

@examlix
Watermarkly
Scanned with OKEN Scanner

rate of 21kg with a relative velocity of 300 m/s.

Concept:

Rocket Propulsion & Newton's Second Law

Solution:

The upward thrust force (F_{thrust}) exerted on the rocket by the ejected gases is given by:

$$F_{thrust} = v_r \cdot \left(\frac{dm}{dt}\right)$$

Substitute the given values:

F_{thrust} = 300 m/s × 2kg/s = 600 N

In gravity-free space, thrust force is the net force acting on the rocket ($F_{net} = F_{thrust}$).

Using Newton's Second Law (F_{net} = m . a):

$$a = \frac{F_{thrust}}{m}$$

Substitute F_{thrust} = 600 N and mass m = 60 kg:

$$a = \frac{600}{60} = 10 m/s^2$$

Final Answer:

Option (3)

27) $\vec{P}(t) = A\left[\hat{i}\cos(kt) - \hat{j}\sin(kt)\right]$

$$\vec{F}(t) = \frac{d}{dt}\left(\vec{P}(t)\right)$$

$$\vec{F}(t) = A\left[-k\hat{i}\sin(kt) - k\hat{j}\cos(kt)\right]$$

$\vec{P}(t).\vec{F}(t) = -KA^2 \sin(kt)\cos(kt) + KA^2 \sin(kt)\cos(kt) = 0$

So, $\vec{P}(t) \perp \vec{F}(t)$

28) **Question Explanation:**

A. Given: M = 1.0 kg, m = 0.05 kg}, n = 10 s^{-1}, g = 9.8 m/s^2.
B. Find: Firing speed (v).

Concept: Newton's Second Law \& Momentum Conservation

Solution:

To keep the disc floating, the upward force (F) of the bullets must balance the disc's weight (Mg): F = Mg The force from elastically rebounding bullets is: $F = n\Delta p = 2nmv$ Equating the two:

$$2nmv = Mg \quad v = \frac{Mg}{2nm} = \frac{1.0 \times 9.8}{2 \times 10 \times 0.05} = 9.8 \text{ m/s}$$

Final Answer: Option (3)

29)

Explain Question : Asking about the force exerted by ball on the wall as shown in the diagram.

Concept : This question is based on second law of motion & impulse.

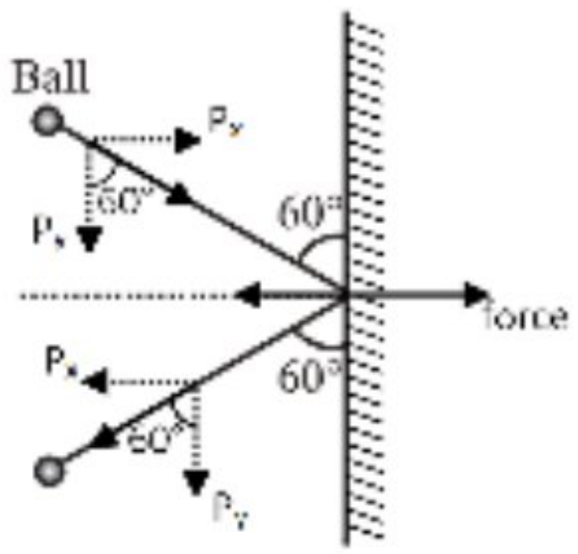
Solution :

@examlix

Watermarkly

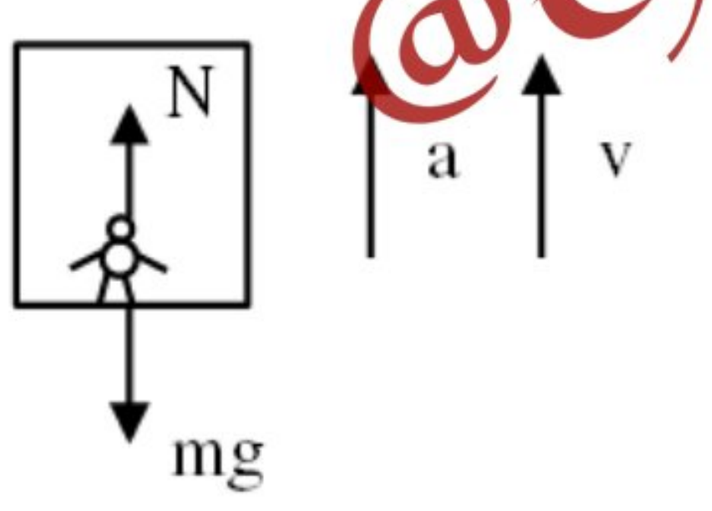
Scanned with OKEN Scanner

NOVA



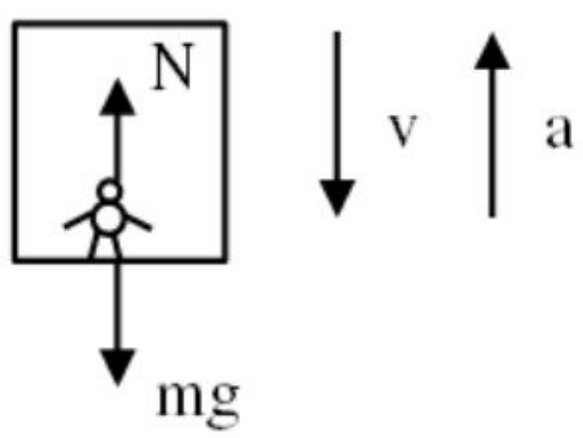
$$\left|\Delta\vec{P}\right| = \left|\left(\vec{P}_f\right) - \left(\vec{P}_i\right)\right|$$
$= 2P_x$
$|\Delta\vec{P}| = 2mv(\sin 60^\circ)$
then from the 2nd law
$$\left|\vec{F}\right| = \frac{\left|\Delta\vec{P}\right|}{\Delta t} = \frac{2mv\sin 60^\circ}{\Delta t}$$
$$\left|\vec{F}\right| = \frac{(2)(3)(100)}{(0.2)} \times \frac{\sqrt{3}}{2}$$
$$\left|\vec{F}\right| = 1500\sqrt{3}N$$
Final answer : Option (1)

30) **Question Explanation :-** To calculate the apparent weight of body.
Concept :- This question is based on Second Law of motion, Frame of Reference
Formula :- F_{net} = mass × acceleration
Solution :- A when lift going up with increasing speed



N – mg = ma
N = mg + ma
N > mg

B. when lift going down with decreasing speed



N – mg = ma
N > mg

@examlix
Watermarkly
Scanned with OKEN Scanner

Final Answer :- Option (3)

31)

Explain Question : Asking about the acceleration of the rocket 50 second after the blast.
Concept : This question is based on Rocket propulsion.
Solution :

$$V_r\frac{dm}{dt} = M'a \quad ...(1)$$

Mass after 50 sec.

$$M' = M - \frac{dm}{dt} \times 50$$

$M' = 1500 - 10 \times 50$

$= 1000$ kg

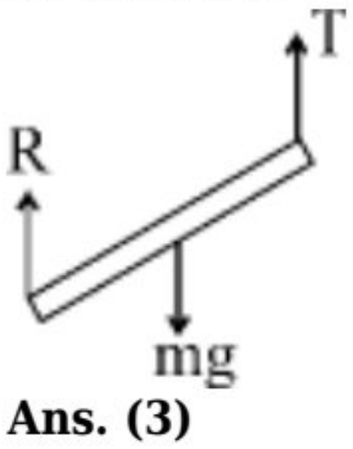
From eq. (1)

$5 \times 10^3 \times 10 = 1000 \times a$

$a = 50\ m/s^2$

Final answer : Option (3)

32) **Explain:** Choose the correct free body diagram (FBD) showing all external force acting on the rod of mass m.
Concept: FBD (Free body diagram)
Formula: Weight W = mg
Calculation:
Weight (mg): Acts vertically downwards from the center of mass.
Normal reaction from the ground (N): Acts vertically upwards perpendicular to the ground.
Tension (T): Acts along the string, which is connected from the top of the rod to the wall.
Correct FBD

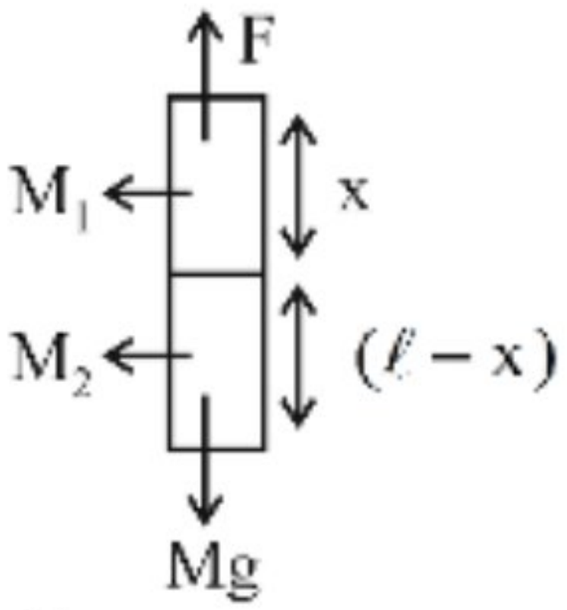


Ans. (3)
Difficulty level: Easy

@examlix
Watermarkly
Scanned with OKEN Scanner

NOVA

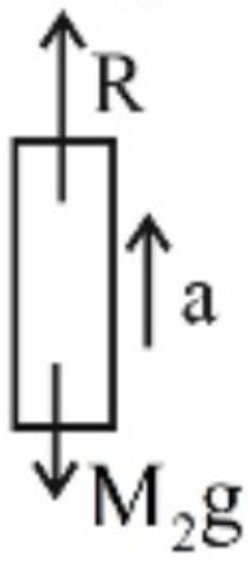
33)



$$M_2 = \frac{M}{\ell}(\ell - x)$$

$$a = \frac{F_{net}}{M}$$

$$a = \frac{F - Mg}{M}$$



$$R - M_{2g} = M_2 a$$

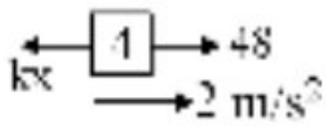
$$R = M_2 (g + a)$$

$$\frac{M}{\ell}(\ell - x)\left(g + \frac{F - Mg}{M}\right)$$

$$R = \frac{M(\ell - x)}{\ell}\left(\frac{(Mg + F - Mg)}{M}\right)$$

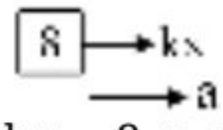
$$R = \frac{F(\ell - x)}{\ell}$$

34)



$48 - kx = 4 \times 2$

$kx = 48 - 8 = 40$ N



$kx = 8 \times a$

$\Rightarrow 40 = 8 \times a$

$$\Rightarrow a = \frac{40}{8}$$

$= 5\ m/s^2$

35)

m_2 is lifted off, if $T \geq m_2 g$

$$\frac{F}{2} \geqslant m_2 g$$

@examlix

Watermarkly

Scanned with OKEN Scanner

$F \geq 2m_2g = 2 \times 8 \times 10$
$= 160$
$2t \geq 160$
$t \geq 80$ sec
$t = 80$ sec

36) **Explanation:**
The question asks which orientation of two applied forces produces the least magnitude of acceleration for the object.

Concept:
According to Newton's second law,
$a = \frac{F_{net}}{m}$
Minimum resultant force produces minimum acceleration.

Solution:
For minimum force, both forces act opposite.
$F_{min} = 5 - 3 = 2N$
The acceleration is least when the 3 N and 5 N forces act in opposite directions.

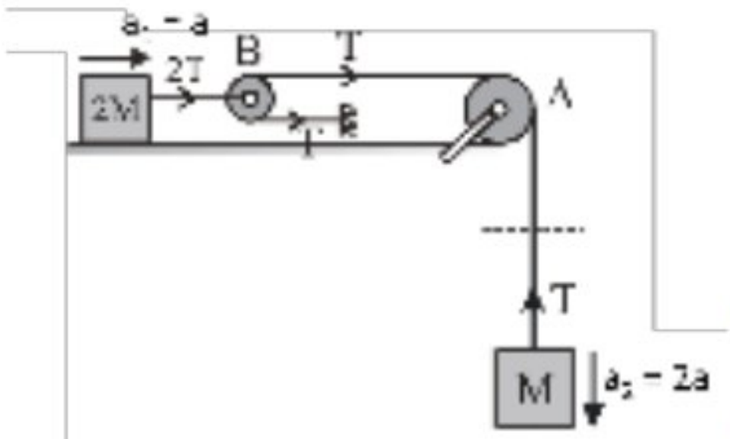
Final Answer: Option (1)

37)

Explain Question:
Question asks for ternion acceleration and reaction force an pulley.

Concept:
Constraint Relation.

Solution:



$T_1a_1 = T_2a_2$
$2Ta_1 = Ta_2$
$a_2 = 2a_1$
Mg – T M (2a) —(1)
2T – 2Ma —(2)

Using (1) & (2)
Mg – Ma = 2Ma
3a = g
$A_1 = a = g/3$
$a_2 = 2g/3$

@examlix
Watermarkly
Scanned with OKEN Scanner

Using (2)

$T = Ma$

$$T = \frac{Mg}{3}$$

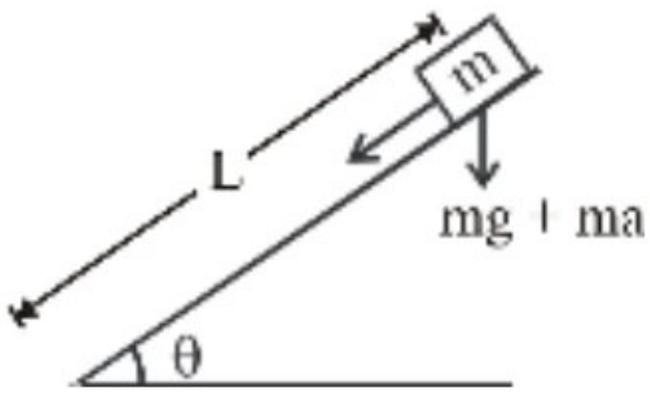
Ternion in string A to B is $\frac{Mg}{3}$

Reaction on pulley $A = \sqrt{2}T$

$$= \frac{\sqrt{2}Mg}{3}$$

Final answer: Option (4)

38)

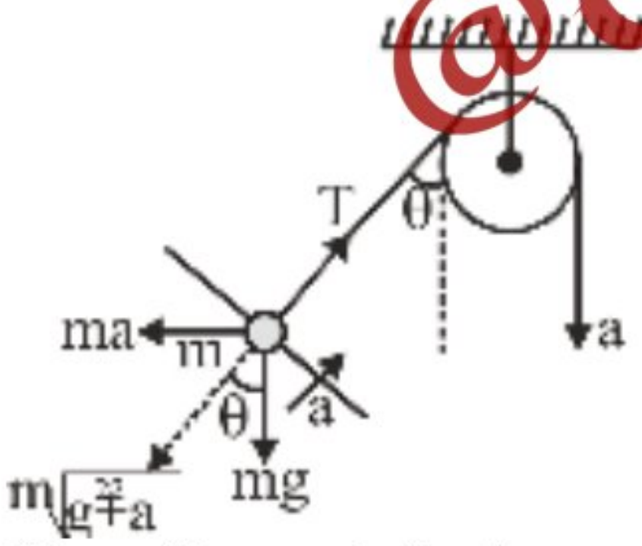


$g_{eff} = (g+a) \sin \theta = a'$

$$s = \frac{1}{2}a' t^2 = L$$

$$t = \sqrt{\frac{2L}{a'}} = \sqrt{\frac{2L}{g \sin \theta + a \sin \theta}}$$

39)



(Force diagram in the frame of the car). Applying Newton's law perpendicular to string mg sin θ = ma cos θ

$$\tan \theta = \frac{a}{g}$$

Applying Newton'a law along string

$$\Rightarrow \quad T - m\sqrt{g^2 + a^2} = ma$$

$$T = m\sqrt{g^2 + a^2} + ma$$

40)

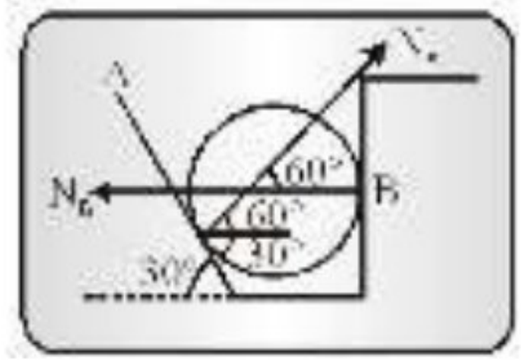
$N_A \sin 60° = 500$

$$N_A = \frac{1000}{\sqrt{3}}$$

@examlix

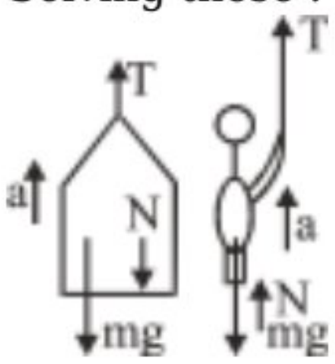
Watermarkly

Scanned with OKEN Scanner



$N_A \cos 60° = N_B \Rightarrow N_B = \frac{500}{\sqrt{3}}$

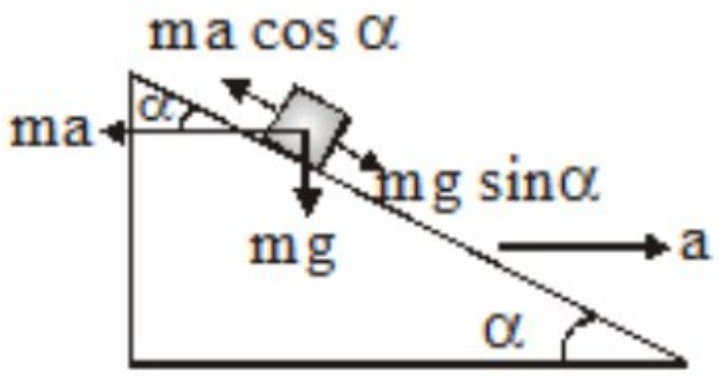
41) $m_1 = 100$ kg, $m_2 = 50$ kg, $a = 5\ ms^{-2}$
$T + N - m_1 g = m_1 a$,
$T - N - m_2 g = m_2 a$
Solving these : $T = 1125$ N



42) **Explain Question:**
Find acceleration of wedge.

Concept:
Newton's 2nd law and equilibrium

Solution:
Let the mass of a block is m.
It will remains stationary if forces acting on it are in equilibrium i.e. $ma\cos\alpha = mg\sin\alpha$
Here ma is the pseudo force on block and mg is the weight.



So $ma\cos\alpha = mg\sin\alpha$
$\Rightarrow a = g\tan\alpha$.

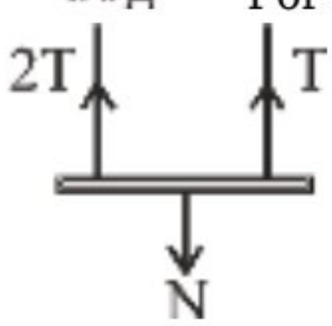
Final answer: Option (2)

43) **Explain the Question :** From the data given and figure shown, we need to find what force the man must apply on the rope to keep the system in the state of equilibrium.
Concept : This question is based on equilibrium (multiple ropes and pulleys)
Solution :

@examlix
Watermarkly
Scanned with OKEN Scanner



For man : N + T = 80 g ...(i)



For plank : 3T = N ...(ii)

$4T = 800$

$T = 200$ N

Final Answer : 1

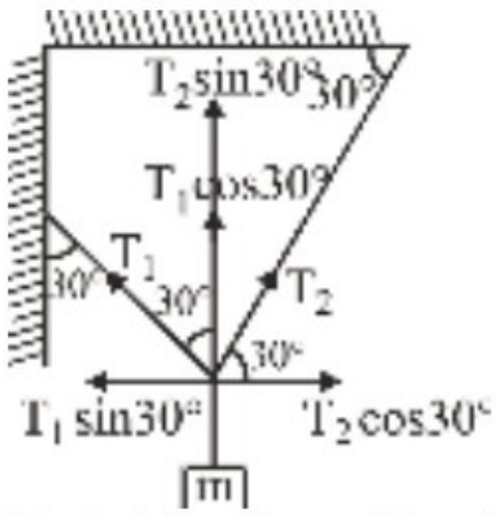
44)

Explain Question : A Block is attached with two strings & in equilibrium we need to find the tension in strings.

Concept : This question is based on equilibrium.

Formula : $\vec{F}_{net} = 0$

Solution :



$T_1 \sin 30 = T_2 \cos 30$ ...(1)

$$\frac{T_1}{2} = \frac{T\sqrt{3}}{2} \Rightarrow T_1 = \sqrt{3}\,T_2$$

$T_2 \sin 30 + T_1 \cos 30 = mg$ ...(2)

$$\frac{T_1}{\sqrt{3}}\frac{1}{2} + \frac{T_1\sqrt{3}}{2} = mg$$

$$\frac{T_1}{2}\left[\frac{1+3}{\sqrt{3}}\right] = mg$$

$$T_1 = \frac{\sqrt{3}mg}{2}$$

$$T_2 = \frac{T_1}{\sqrt{3}} = \frac{mg}{2}$$

@examlix

Watermarkly

Scanned with OKEN Scanner

Final Answer : option (1)

45) $\frac{6}{Mo_2} = \frac{PV_0}{R(400)}$ ...(1)

After leakage

$\frac{m}{Mo_2} = \frac{(P/2)V_0}{R(300)}$ ...(2)

from (1) & (2)

$\frac{6}{m} = \frac{600}{400}$

$\Rightarrow m = 4g$

Gas leaked = 6 – 4 = 2g

CHEMISTRY

46) Number of molecules of water in

$1mL = \frac{1}{18} \times 6.02 \times 10^{23}$

$= 3.346 \times 10^{22}$

No. of water molecules in one drop = $\frac{3.346 \times 10^{22}}{20} = 1.67 \times 10^{21}$

47) No. of molecules = $\frac{wt.}{mol.wt.} \times N_A$

$= \frac{0.2}{44} \times 6.023 \times 10^{23}$

$= 2.74 \times 10^{21}$ molecules

The residual CO_2 molecules after removal of 10^{21} molecules

$= 2.74 \times 10^{21} - 10^{21}$

$= 1.74 \times 10^{21}$ molecules

No. of moles = $\frac{\text{No. of molecules}}{N_A}$

$= \frac{1.74 \times 10^{21}}{6.023 \times 10^{23}} = 2.88 \times 10^{-3}$

48)

Correct Answer is Option (1).

49)

Correct Answer is Option (4).

50)

51)

@examlix

Watermarkly

Scanned with OKEN Scanner

Correct Answer is Option (3).

52)

Correct Answer is Option (3).

53) **Explanation the Question :**
We need to calculate the volume of CO_2 at STP released when 10 g of $CaCO_3$ (90% pure) decomposes.
Reaction:
$CaCO_3 \rightarrow CaO + CO_2$
Concept : This question is based on
1 mol $CaCO_3 \rightarrow$ 1 mol CO_2 (22.4 L at STP)
Purity must be considered: pure mass = given mass × % purity
Solution:
Pure mass of $CaCO_3$ = 10 × (90/100) = 9 g
Moles of $CaCO_3$ = 9 ÷ 100 = 0.09 mol
Moles of CO_2 formed = 0.09 mol (1:1 ratio)
Volume of CO_2 at STP = 0.09 × 22.4 = 2.016 L
Final Answer : option (1)

54) I oxide = MO

	M	:	O
	$\frac{77.78}{M}$	:	$\frac{22.22}{16}$
mol :			= 1 : 1

⇒ M = 56
II oxide = ?

	M	:	O
	$\frac{70}{56}$	:	$\frac{30}{16}$
mol :			

= 2 : 3 ⇒ M_2O_3

55)

Weight of impure limestone = 20 g

Weight of pure limestone ($CaCO_3$) = 20% of 20 g

$$\frac{20}{100} \times 20$$

= 4g

$$n_{CaCO_3} = \frac{4}{100} = 0.04$$

$CaCO_3 \rightarrow CaO + CO_2$

n=0.04 n=0.04

$n_{CO_2} = 0.04$

$W_{CO_2} = 0.04 \times 44$

= 1.76 g

@examlix
Watermarkly
Scanned with OKEN Scanner

56)

Correct Answer is Option (2).

57)

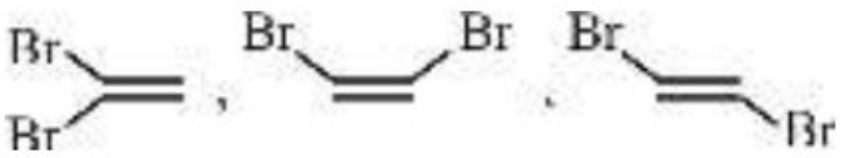
Correct Answer is Option (2).

58)

Correct Answer is Option (1).

59)

Position of Cl and F is changed

60) 

61) **Question Explanation:** The objective is to identify which pair of organic compounds is correctly classified by the specific type of structural isomerism mentioned in the options.

Concept: Isomerism, Position isomers, Molecular formula, Functional isomers, Carbon chain length.

Solution:
We must first verify if the pairs have the same molecular formula:
Option 1: Both are methylcyclopropene (C_4H_6). The double bond is at position 1 in the first and position 2 (or 3 relative to methyl) in the second. These are Position isomers. Correct.
Option 2: Propan-1-ol (C_3H_8O) and Methoxymethane (C_2H_6O). Carbon counts differ (3 vs 2). Not isomers.
Option 3: Butan-2-one (C_4H_8O) and Propanal (C_3H_6O). Carbon counts differ (4 vs 3). Not isomers.
Option 4: 1-methoxypropane and 2-methoxypropane. Both are $C_4H_{10}O$. The position of the methoxy group changes, making them Position isomers, not chain isomers.

Final Answer: Option (1)

62)

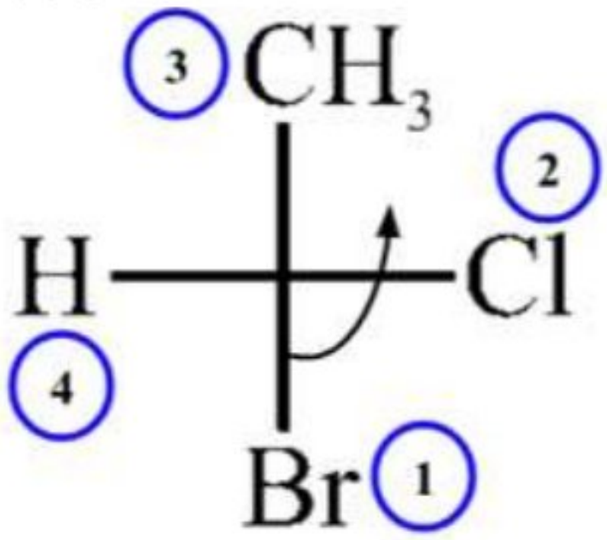
All are correct

63)

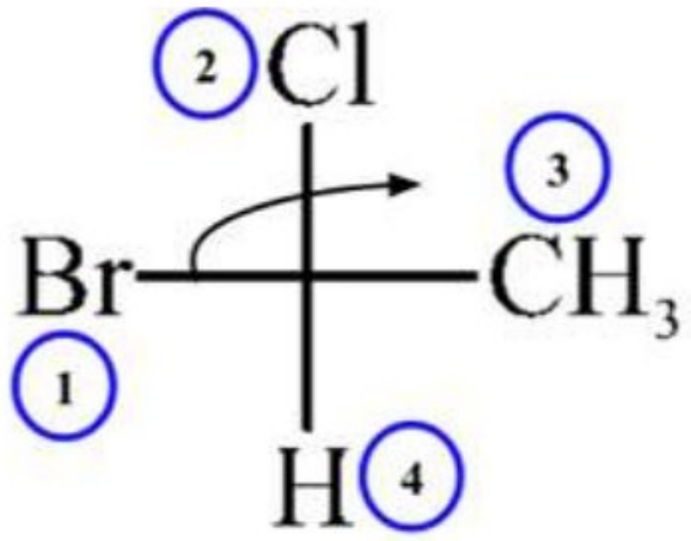
Question Explanation: Identify the stereochemical relationship (Enantiomers, Diastereomers, Meso, or Homomers) between the two given chiral structures.

@examlix
Watermarkly
Scanned with OKEN Scanner

Concept Used: Stereoisomerism, R/S Configuration, Cahn-Ingold-Prelog Rules, Homomers
Solution :



4th Priority on Horizontal
Anticlockwise
"R"



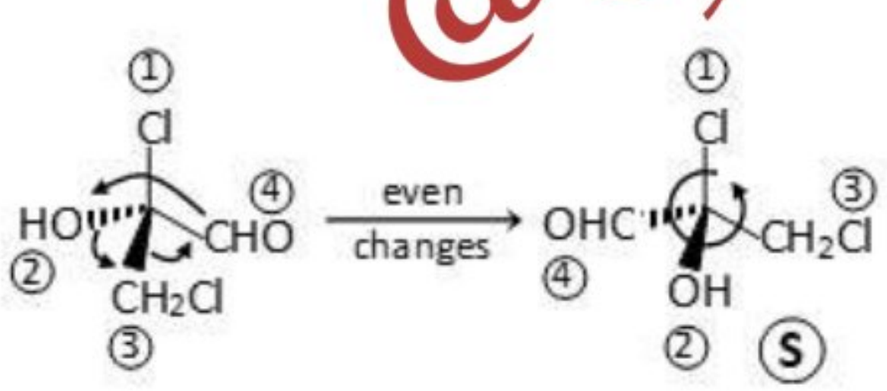
4th Priority on Vertical
Clockwise
"R"

Conclusion: Both structures have the **(R)-configuration**. Identical configurations imply they represent the same molecule (Homomers).

Final Answer: Option (4) Homomers

64)

Explanation the Question:
The R-S configuration of the given compound will be.
Concept:
This question is based on R-S configuration.
Solution:



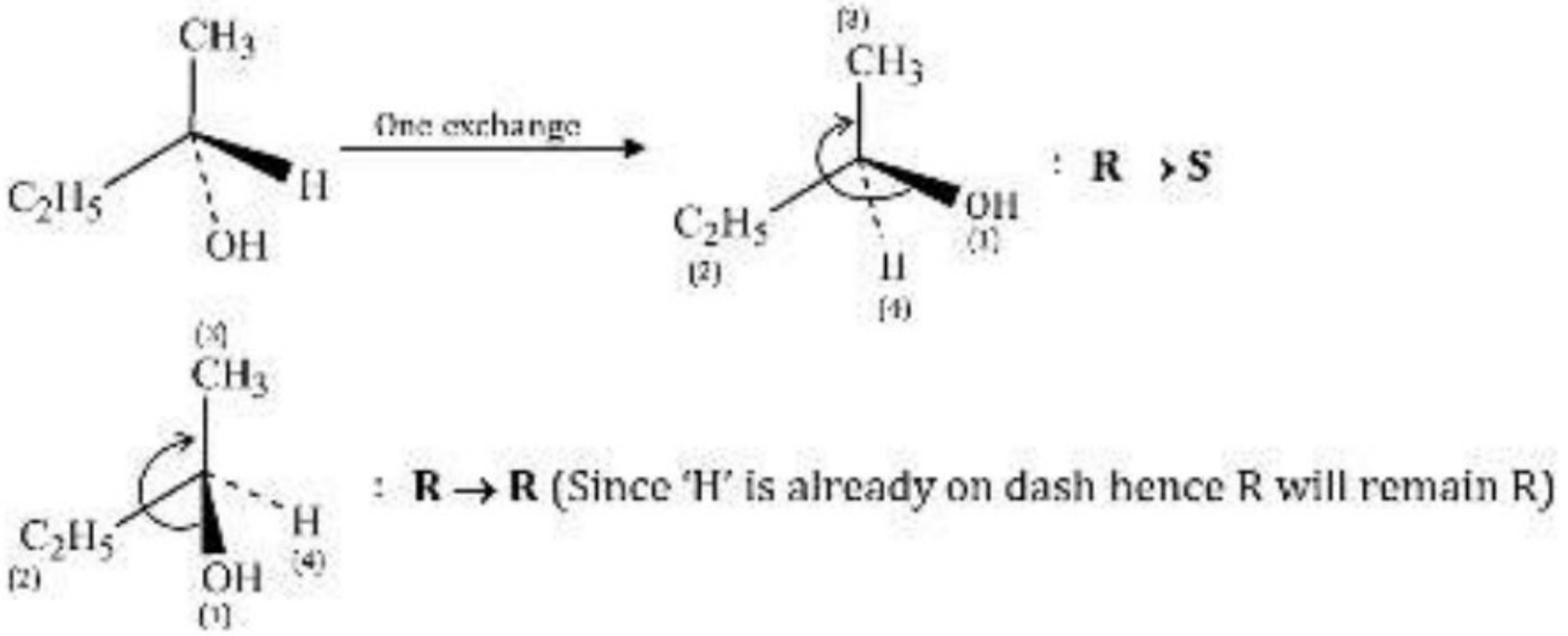
65) **Question Explanation:** The user is verifying the relationship between two structures of Butan-2-ol where the positions of the –H and –OH groups are swapped at the chiral center.

Concept: Chiral center, Wedge-dash notation, Single exchange rule, Enantiomers.

Solution:
The Rule of Interchange: In a 3D projection, a **single exchange** (odd number of swaps) between any two groups attached to a chiral center results in the opposite configuration (the enantiomer).

@examlix
Watermarkly
Scanned with OKEN Scanner

NOVA



Since both the structure are non-superimposable mirror image of each other, hence are Enantiomers .

Final Answer: Option (1)

66)

Question Explanation: The question asks which conformation of ethylene glycol is the most stable.

Concept: Conformational analysis, intramolecular hydrogen bonding, gauche effect, anti conformation

Solution:

For ethylene glycol, HO–CH_2–CH_2–OH, the **gauche conformation** is more stable than the anti form because the two –OH groups can form **intramolecular hydrogen bonding**. **Gauche form**: The two –OH groups are close enough to form intramolecular H-bonding, which provides additional stabilization.

Conformation	Stability
Fully eclipsed	Least stable
Partially eclipsed	Less stable
Anti	Less stable than gauche
Gauche	**Most Stable**

HO–CH_2–CH_2–OH → **Gauche form → Intramolecular H-bonding → Higher stability**

Final Answer: Option (1).

67)

Question Explanation : Determine which of the given cyclic or acyclic compounds does not exhibit geometrical isomerism.

@examlix
Watermarkly
Scanned with OKEN Scanner

Concept:

A. **Cycloalkenes:** Double bonds inside small or normal ring systems (rings with fewer than 8 carbon atoms, such as cyclopentene and cyclohexene) cannot exhibit geometrical isomerism across the double bond because the trans-configuration is extremely ring-strained and unstable. Geometrical isomerism around a double bond in a ring is only possible for large rings (≥ 8 carbon atoms).
B. **Ring Substituents:** 1, 2-disubstituted cycloalkanes or substituted rings can show cis/trans isomerism across the ring carbons.

Solution:

A. **Option (1):** Contains a 10-membered ring with a double bond. Large rings (≥ 8 carbons) are flexible enough to accommodate both cis and trans double bonds. ⟹ Shows geometrical isomerism.
B. **Option (2):** A substituted cyclohexene ring with adjacent 1, 2disubstituted sp3sp^3 carbons (Cl and –Br). These ring carbons can have cis and trans spatial arrangements relative to the ring plane. ⟹ Shows geometrical isomerism.
C. **Option (3):** A 5-membered ring (cyclopentene) with a double bond. Due to intense ring strain, small rings (< 8 carbons) can exist only in the cis-configuration at the double bond. The trans-isomer does not exist at room temperature, so it cannot show geometrical isomerism across the double bond.
D. **Option (4):** An acyclic alkene (3-methylhex-3-ene). Both double-bonded carbons have two different groups attached (–Et, –Me on one side and –Me, –H on the other), enabling E/Z isomerism. ⟹ Shows geometrical isomerism.

Final Answer : Option (3)

@examlix

68)

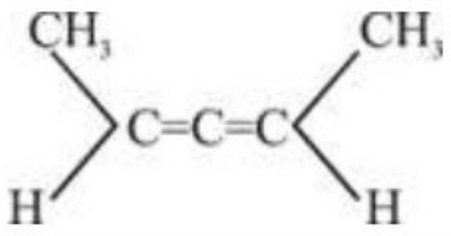
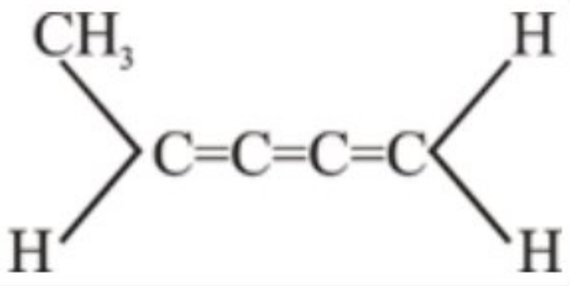
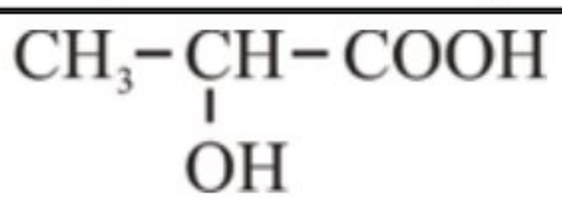
Explaining the Question : The goal is to identify which cyclohexane derivative cannot exist as cis/trans isomers (geometrical isomers)
Concept : This question is based on For geometrical isomerism in a ring, at least two different sp^3 carbons must each be attached to two different groups. If any carbon holds two identical groups (e.g., two chlorines), it cannot create a distinct 'cis' of "trans" configuration relative to another carbon.
Solution :
• Options 1, 2 and 3 : The chlorine and Hydrogen atoms are attached to different sp^3 carbons. Each carbon has a distinct "up" or down" position, allowing for various cis/trans combinations.
• Options 4 : The first carbon is bonded to two identical Chlorine atoms. Because these groups are the same, there is no way to designate one as "cis" and the other as "trans" relative to the chlorine at position 4. It forms only one unique compound.
Final Answer : 4

69)

Explain the Question :- Given compound can show geometrical and optical isomerism
Concept :-This question is based on Stereoisomerism
Solution :-

Watermarkly

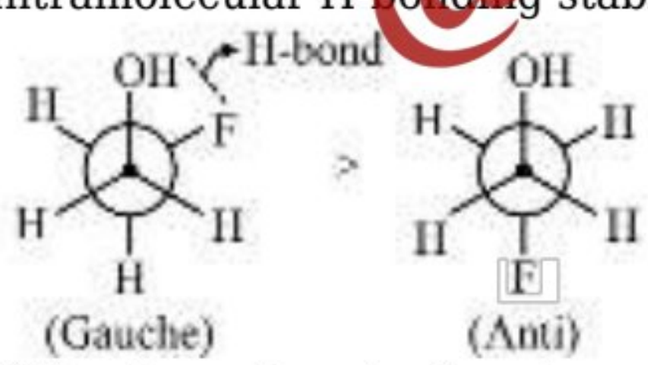
Scanned with OKEN Scanner

	Given compound	Geometrical isomerism	Optical isomerism
1		✗	✓
2		✗	✗
3		✓	✓
4		✗	✓

Final answer :- 3

70)

Question Explanation: 'A' is more stable than 'B' if 'X' is
Concept: Stability of conformers.
Solution :
Intramolecular H-bonding stabilises the gauche form.



If X = F, so, Gauche form is more stable than anti form due to intramolecular H-bonding.
Final Answer: Option (2)

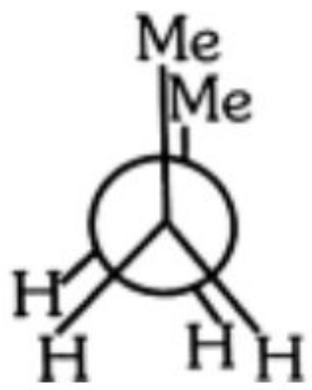
71)

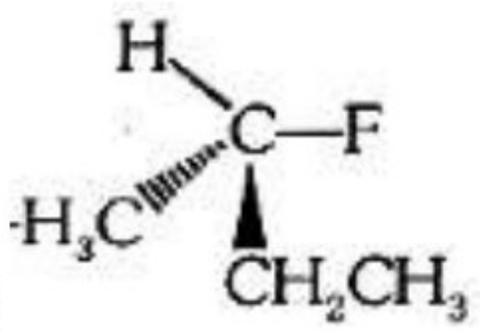
Correct Answer is Option (1).

72)

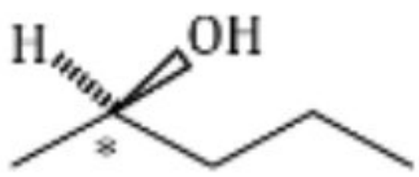
Correct Answer is Option (3).

@examlix
Watermarkly
Scanned with OKEN Scanner

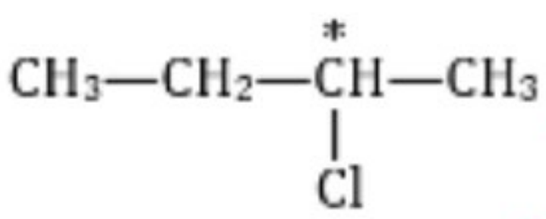
73) 

74) 

75) **Explaining :** Total number of optically active compounds from the following is :-
Concept : This question is based on chiral molecule.
Solution :



Chiral centre



Chiral centre



Chiral centre

Final Answer : (2)

76)

Correct Answer is Option (3).

77) IP of mono anion is calculated by
EA so, answer $Cl^- > F^-$

78) **Question Explanation:**
Identify true or false

Concept:
This question is based on electronic configuration.

Solution:
(I) F [1]. The order of energy for a multi-electron atom like Mn is 3s < 3p < 3d [1]. The energies are equal only for a hydrogen atom or hydrogen-like ions.
(II) F [1]. The 5th period of the long-form periodic table contains 18 elements [1].
(III) T [1]. The ground state electronic configuration of Zr(Z = 40) is $[Kr]5s^24d^2$. For the Zr^{+2} ion, the configuration becomes $[Kr]4d^2$, which has $n = 2$ unpaired electrons [1, 2]. The magnetic moment is calculated using the spin-only formula $\mu_m = \sqrt{n(n+2)}$ BM, resulting in $\mu_m = \sqrt{2(2+2)} = \sqrt{8}$ BM [1, 2].
(IV) F [1]. The general valence shell electronic configuration for *f*-block elements is $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$, where the *d*-orbital can have 0 or 1 electron, not a full 10 electrons [1, 2].

@examlix
Watermarkly
Scanned with OKEN Scanner

Final Answer:
The correct option is (3)

79)

Correct Answer is Option (1).

80) **Question Explanation :** Match the first Ionization Energy (IE1) values (in kJ mol^{-1}) to the elements Be, Ne, He, N.

Concept: This question is based on factors affecting IE.

Solution:

$$IE \propto \frac{Z_{eff}}{\text{number of shell}}$$

Be < N < Ne < He

↓ ↓ ↓

Z_{eff} $_Z$eff Number of shell

Final Answer: Option (3)

81)

Correct Answer is Option (1).

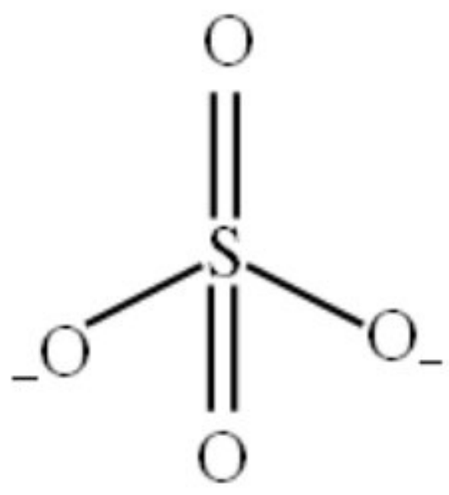
82)

@examlix

Explaining the Question :- Question is asking to choose species having bond order 1.5

Concept:- This question is based on MOT and Resonance

Solution :- (1) SO_4^{2-}

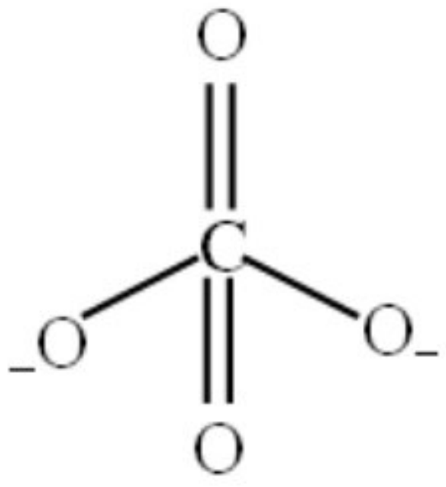


$$B.O = \frac{6}{4} = 1.5$$

(2) O_2 $B.O = \frac{1}{2}(Nb - Na) = \frac{1}{2}(10 - 7) = 1.5$

Watermarkly

Scanned with OKEN Scanner

(3) CO_3 

$\frac{4}{3} = 1.33$

SO_4^{2-} and O_4^{-} both have bond order 1.5

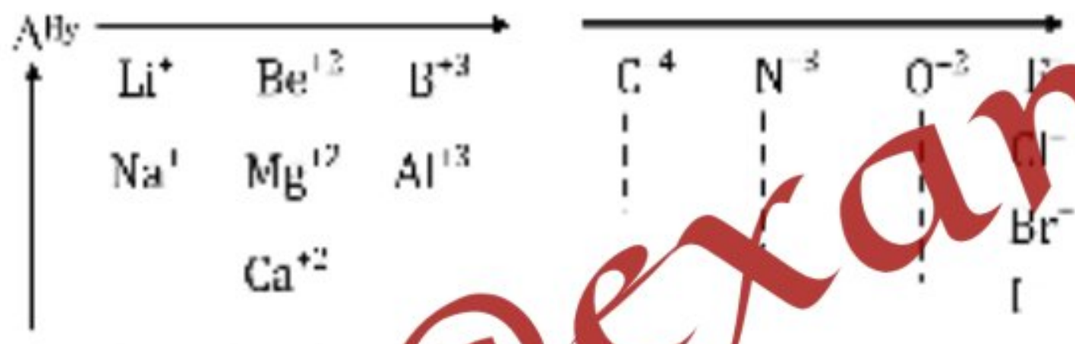
Final Answer :- 4

83)

Explanation of question: Question is asking about incorrect radius order.

Concept: This question is based on Hydration.

Solution: Higher extent of hydration depends upon higher charge density.
Periodicity :-



 Hydrated radius $Cl^- > I^-$
Rest options are correct
Down the group hydrated size decreased due to lower change density
 Cl^- (ag) > I^- (aq) (in size)

Final Answer: The correct option is (3)

84) **Question Explanation:** Identify the incorrect covalent character order.
Concept: Factors affecting covalent character.
Solution: Covalency increases with higher cationic charge and smaller size. Transition metals/pseudo noble gas ions are more covalent than s-block ions of similar size.
Zn^{2+} is a transition metal ion (pseudo noble gas) and is much more polarising than s-block Ca^{2+}. The correct order is $ZnCl_2 > CaCl_2$. The given order is **incorrect**.
Final Answer: Option (2)

85)

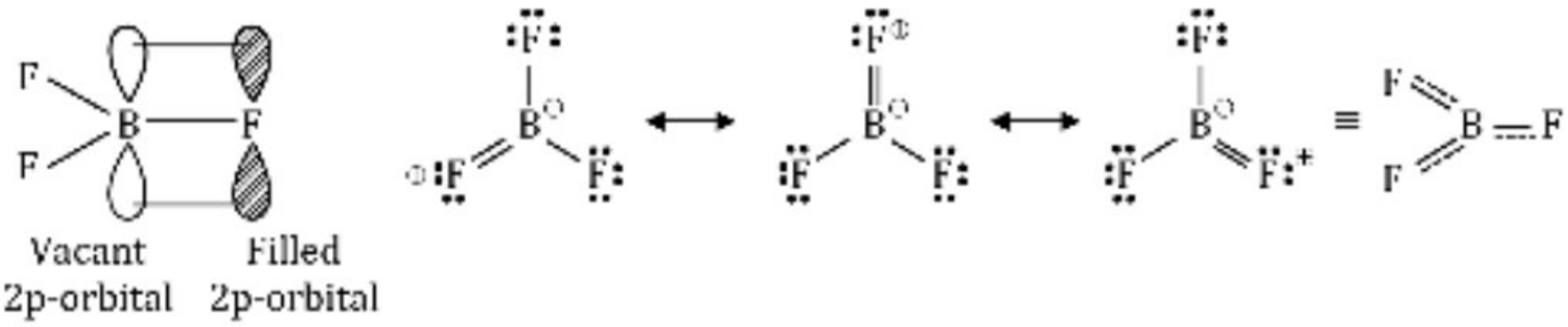
Question Explanation: Verifying structural and bonding properties of the BF_3 molecule.

Concept: Properties of BF_3

@examlix
Watermarkly
Scanned with OKEN Scanner

Solution: All given statements for BF_3 are correct:

A. **Hybridization:** The Boron atom is **sp^2 hybridized**, leading to a trigonal planar geometry.
B. **Back Bonding:** There is $p\pi - p\pi$ **back bonding** where Fluorine (donor) shares its 2p lone pair with the vacant 2p orbital of Boron (acceptor).
C. **Bond Length:** The observed B—F bond length is **shorter than expected** because the back bonding introduces partial double-bond character.



Decrease in B–F bond length is due to delocalised pπ–pπ bonding between filled p-orbital of F atom and vacant p-orbital of B atom.

Final Answer: All of these

86)

Question Explanation: Identify the orbital diagram that does not follow the Aufbau principle.

Concept: Lower energy orbitals fill first

Solution:
According to the Aufbau principle, the 2s orbital must be completely filled before electrons enter the 2p orbitals.

Checking the options:

A. **Option (1):** $2s^2 2p^4$ Correct
B. **Option (2):** $2s^1 2p^4$ Incorrect, because 2p has electrons while 2s is not completely filled.
C. **Option (3):** $2s^2 2p^3$ Correct
D. **Option (4):** $2s^2 2p^5$ Correct

Therefore, Option (2) violates the Aufbau principle.

Final Answer: Option (2)

87)

Correct Answer is Option (4).

88)

Explain the question: Find the entity with the maximum electron capacity.

Concept: Electron capacity rules: shell = $2n^2$; orbital = 2 (Pauli Principle).

@examlix
Watermarkly
Scanned with OKEN Scanner

Solution:
The maximum electron capacity for each given condition is calculated based on quantum number rules:

A. **Option 1 (n = 4):** Represents an entire principal shell. The total capacity is determined by $2n^2 = 2 \times 4^2 = 32$ electrons.
B. **Option 2 (Subshells with n + ℓ = 5):** Possible combinations are 5s(n = 5, ℓ = 0), 4p(n = 4, ℓ = 1) and 3d(n = 3, ℓ = 2). The respective capacities are 2, 6 and 10, totaling 2 + 6 + 10 = **18** electrons.
C. **Option 3 (n = 3, ℓ = 2, m = 0):** Since all three spatial quantum numbers are fixed, this describes a single specific orbital ($3d_{z^2}$). Any single orbital can hold a maximum of 2 electrons with opposite spins.
D. **Option 4 ($2p_z$):** Similar to Option 3, this refers to one specific orbital within the 2p subshell, limiting its capacity to 2 electrons. Comparing the results, the n = 4 shell has the largest capacity.

Final Answer: Option (1)

89) Any orbital can accommodate only 2 electrons with opposite spins

90)

Correct Answer is Option (1).

BIOLOGY

@examlix

91)

Explain Question – The question is asking about to identify the correct labels in a cell membrane diagram.
Concept – Cell membrane
Solution – A: Sugar B: Protein C: Lipid bilayer D: Cholesterol
Final answer – Option 3

92)

Explain Question : The question shows a diagram of the Golgi apparatus and asks you to correctly identify the labeled parts (A, B, C, D) with their proper names and functions based on structure and orientation.
Concept : This question is based on the Golgi apparatus is a cell organelle involved in the packaging and transport of proteins and lipids.
Explanation : In the given diagram of the Golgi apparatus:
A → Cisternae (flattened membrane-bound sacs)
B → Vesicle (small membrane-bound structure for transport)
C → Trans face (maturing face; shipping side)
D → Cis face (forming face; receiving side from ER)
Final answer : (1) Cisternae | Vesicle | trans face | cis face

Watermarkly

Scanned with OKEN Scanner

93) NCERT XI, Page # 21

94) NCERT Page # 14

95)

Correct Answer is Option (3).

96)

Correct Answer is Option (2).

97)

Correct Answer is Option (2).

98)

Solution :
The correct answer is
4. They play an important role in movement.
Pili are involved in conjugation and attachment, not in movement. Flagella are responsible for bacterial movement.

99) **Solution :** The capsomere is a subunit of the capsid, an outer covering of protein that protects the genetic material of a virus. Capsomeres self-assemble to form the capsid.
Final Answer :: (4)

100) **Solution :** "Late blight of potato" Disease is caused by a member of Phycomycetes.
Final Answer : (1)

101) **Solution :** Albugo is the parasitic fungi on mustard and belongs to the class phycomycetes and kingdom fungi.
Final Answer :: (1)

102) NCERT (19, 21, 23, 24)

103)

Correct Answer is Option (2).

104)

Correct Answer is Option (1).

@examlix
Watermarkly
Scanned with OKEN Scanner

105)

Correct Answer is Option (3).

106) NCERT (XIth) Pg. # 27

107)

Correct Answer is Option (4).

108)

Explain Question : The question asks for the specific type of flagellation present on the male gametes (antherozoids) of bryophytes
Concept : This question is based on Bryophyta, General Characters And Life Cycle
Explanation : In bryophytes male sex organ is called the antheridium. They produce biflagellate antherozoids
Final answer :4

109)

Explain Question : The question asks us to match three categories of plants from Column-I with their corresponding examples in Column-II

Concept : This question is based on classification of the plant kingdom

Solution :

A. Non-vascular plants:
These plants lack specialized vascular tissues (xylem and phloem) for transporting water and nutrients. Examples include Algae and Bryophytes. Therefore, A matches with (iii).

B. Vascular plants:
These plants possess a well-developed vascular system. Pteridophytes (like ferns) are examples of vascular plants. Therefore, B matches with (ii).

C. Seeded plants:
These plants reproduce through seeds. Gymnosperms (like conifers) and Angiosperms (flowering plants) are examples of seeded plants. Therefore, C matches with (i).

Final Answer : option (1)

110)

Explain Question : The question asks to identify a plant based on a provided diagram and then determine its name, class, and the ploidy of its rhizome.
Concept : This question is based on Sphenopsida
Explanation :
(a) Name of plant:

@examlix
Watermarkly
Scanned with OKEN Scanner

The diagram clearly depicts the characteristic features of Equisetum, including its jointed stems and whorled branches.

(b) Class of plant:

Equisetum is a member of the class Sphenopsida, which is a group of seedless vascular plants.

(c) Ploidy of rhizome:

The rhizome of Equisetum is part of the sporophyte generation, which is diploid (2n).

Final answer : 4

111) NCERT (XI) P.No. 30

112) NCERT-XI, Pg.# 32

113) A protonema is a key structure in the life cycle of bryophytes (such as mosses). It is part of the gametophyte generation and develops from a spore after it germinates. Here's what makes it distinctive:

• **Creeping :** The protonema is typically a creeping or branching structure that grows along the surface.

• **Green :** The protonema is usually green because it contains chlorophyll and can carry out photosynthesis.

• **Filamentous :** It is composed of filaments or thread-like structures, and it looks somewhat similar to fungal mycelium.

• **Gamete-producing :** The protonema will give rise to gametophytes (the male and female reproductive organs) in the next stages of development.

@examix

114)

Answer:

2. Transport of molecules across itSolution:

One of the most important functions of the plasma membrane is the transport of molecules across it. The plasma membrane regulates the entry and exit of various substances into and out of the cell, which is essential for maintaining homeostasis. This includes processes like passive diffusion, active transport, and facilitated diffusion. While exocytosis and detoxification also occur in cells, they are not the primary function of the plasma membrane. The nuclear membrane formation is unrelated to the plasma membrane.

115)

116)

Correct Answer is Option (1).

117) Schleiden and schwann

118) Correct Answer is Option (4).

Watermarkly

Scanned with OKEN Scanner

NCERT Pg # 34

119)

Correct Answer is Option (1).

120)

NCERT, XI, Page No.- 96

121)

Correct Answer is Option (4).

122)

Correct Answer is Option (3).

123)

Old NCERT-XI, Pg # 4

124)

Correct Answer is Option (1).

125)

Correct Answer is Option (2).

126)

Explanation :- You are asked to select correct statements
Concept :- Monera and protista
Solution :-

Statement	correct	Explanation
(I)	yes	All prokaryotes - monera
(II)	yes	All unicellulareukaryotes - protista
(III)	yes	ehlamydomonas schlorella have cell walls
(IV)	yes	Anoeba and parameium lack cell walls
(V)	yes	All grouped in protista

Answer :-

127)

Correct Answer is Option (2).

@examlix
Watermarkly
Scanned with OKEN Scanner

128)

Concept :- Fungi and Animalia.
Solution :- Fungi- Only heterotrophic (mainly saprophytic)
Animalia - Only heterotrophic (Ingestion food)
Answer :- 3

129)

Solution:

The correct answer is:

4. All of the above

130)

Correct Answer is Option (1).

131)

Correct Answer is Option (1).

132) **Solution :** Methanogens → marshy areas Halophiles → salty Thermoacidophiles → hot springs
Final Answer : (1)

@examlix

133)

Explain Question : The bacterial cell is very simple in A but very complex in B. You are to select the correct pair (A and B) that completes this statement accurately

Concept : This question is based on Kingdom Monera

Solution : The correct answer is 4. A = Structure, B = Behaviour
Explanation: Bacterial structure is very simple, they are very complex in behaviour Bacteria are structurally simple:
–No nucleus or membrane-bound organelles
Prokaryotic cell structure
So, A = Structure

Despite their simplicity, bacteria show complex behavior:
– Adaptation, resistance, gene exchange (conjugation, transformation)
So, B = Behaviour

Final Answer : option (4) A = Structure, B = Behaviour

134) **Solution:**

Watermarkly

Scanned with OKEN Scanner

A (Bacteriophage):
Bacteriophages are virsuses that infect bacteria.
B (Agaricus):
Agaricus is a genus of mushroom-forming fungi, including edible species like the common mushroom.
C (Aspergillus):
Aspergillus is a genus of mold species found worldwide.
D (Tobacco mosaic virus):
The tobacco mosaic virus is a rod-shaped virus that infects plants, notably tobacco.

Correct option is (2)

135)

Question Explanation: Asking to Identify the given figure and select the correct statements regarding it ?

Concept: Euglena

Solution: The figure is Euglena (elongated body with a flagellum).



@examlix

(ii) Euglena has one functional flagellum → Correct

(iii) Euglena contains chlorophyll pigments similar to higher plants → Correct

(i) Incorrect → it is not Paramecium

(iv) Incorrect → Paramecium is ciliated, but this organism is flagellated

Final answer: 4(ii) and (iii).

136)

Correct Answer is Option (4).

137) NCERT-XI- 271,272

138)

Correct Answer is Option (3).

139)

Correct Answer is Option (2).

Watermarkly

Scanned with OKEN Scanner

140)

Correct Answer is Option (1).

141)

Correct Answer is Option (3).

142)

Correct Answer is Option (2).

143)

Correct Answer is Option (2).

144)

Correct Answer is Option (2).

145)

NCERT-XI, Pg. # 184

146)

Correct Answer is Option (4).

147)

NCERT - XI, Page No. 275, Para - 17.5

148) NCERT XII, Page no. 269

149)

Correct Answer is Option (4).

150)

Correct Answer is Option (4).

151)

Correct Answer is Option (3).

@examlix

Watermarkly

Scanned with OKEN Scanner

152)

Correct Answer is Option (2).

153)

Correct Answer is Option (4).

154)

Correct Answer is Option (1).

155) NCERT-XI, Pg. # 272

156) NCERT-XI, Pg. # 274

157) NCERT, Pg. # 270

158) NCERT Pg. # 275 (17.6)

159)

Correct Answer is Option (2).

@examlix

160) Notochord is replaced by vertebral column in adults.

161) **Explaining :**
Concept :
Solution : Statements b, c, and d are incorrect. Non-chordates possess ventral, solid and double nerve cord. Heart in non-chordates is dorsal but not always present. Water vascular system is present in echinoderms that helps in locomotion, respiration and capture and transport of food but does not help in reproduction.
Final Answer : (3)

162) Animals like annelids, arthropods, etc., where the body can be divided into identical left and right halves in only one plane, exhibit bilateral symmetry.

163) Mammals are commonly viviparous, and presence of mammary glands is the unique feature of mammals.

164)

Correct Answer is Option (4).

Watermarkly

Scanned with OKEN Scanner

165)

Correct Answer is Option (3).

166)

Correct Answer is Option (3).

167)

Correct Answer is Option (1).

168) **Explaining :**
Concept :
Solution : Gorgonia (Sea fan).
Final Answer : (4)

169) NCERT- XI, Page No.- 44

170)

Correct Answer is Option (4).

171)

Correct Answer is Option (1).

172)

Correct Answer is Option (1).

173)

Correct Answer is Option (2).

174)

NCERT Page # 106

175)

Correct Answer is Option (1).

176)

@examlix

Watermarkly

Scanned with OKEN Scanner

Concept:

A. Helices in proteins can exist in both left-handed and right-handed forms.
B. The tertiary structure refers to the overall 3D shape, and is indeed crucial for many proteins' biological functions.

Explanation:

A. Statement 3 is incorrect: Both left- and right-handed helices are observed in proteins. The most common helices are right-handed alpha helices.

Final Answer: Option 3: Only left helices are observed in proteins

177)

The correct answer is : 2. Rises at first and reaches a maximum velocity

Explanation : As the substrate concentration increases, the velocity of the enzyme-catalyzed reaction initially increases because more substrate molecules are available for enzyme binding. However, once all enzyme active sites are occupied, the reaction reaches a maximum velocity (V_max), beyond which further increases in substrate concentration will not increase the reaction rate. This is due to enzyme saturation.

178) NCERT PG. No 147

179)

Question Explanation: The question asks to identify the correct conclusion drawn from the bell-shaped curve representing the effect of pH on enzyme activity.

Concept: Effect of pH on Enzyme Activity (Optimum pH)

Solution: Enzymes function efficiently within a narrow pH range. Each enzyme exhibits its maximum catalytic velocity at a specific value called the optimum pH. The classic graph plotting enzyme activity against pH forms a bell-shaped curve, indicating that the enzyme shows its highest activity at this particular optimum pH. Both below and above this optimal value, the enzymatic activity steadily declines due to alterations in the ionization state of active site amino acid residues or structural denaturation.

Final answer: Option - 1 (Enzyme show its highest activity at a particular pH, both below and above this pH, activity declines)

180)

Correct Answer is Option (2).

@examlix
Watermarkly
Scanned with OKEN Scanner